

Vehicle Detection for Autonomous Driving: A Review of Algorithms and Datasets

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Abstract—Nowadays, vehicles with a high level of automation are being driven everywhere. With the apparent success of autonomous driving technology, we keep working to achieve fully autonomous vehicles on roads. Efficient and accurate vehicle detection is one of the essential tasks in the environment perception of an autonomous vehicle. Therefore, numerous algorithms for vehicle detection have been developed. However, their strengths in terms of performance have not been deeply assessed or highlighted yet. This work comprehensively reviews the existing methods and datasets for vehicle detection considering their performances and applications in the field of autonomous driving. First, we briefly describe tasks, evaluation criteria, and existing public datasets for vehicle detection in autonomous driving. Second, we provide a rigorous review of both classical and latest vehicle detection methods, including machine vision-based, mmWave radar-based, LiDAR-based, and sensor fusion-based methods. Finally, we analyze the pertinent challenges of autonomous vehicles and provide recommendations for future works concerning vehicle detection. The present review covers over 300 research works and aims to help researchers interested in autonomous driving, especially in vehicle detection.

Index Terms—Autonomous driving, camera, LiDAR, mmWave radar, sensor fusion, vehicle detection.

I. INTRODUCTION

VEHICLES that can drive autonomously or provide relevant information to the driver for safety, assistance, and comfort are called intelligent vehicles (IVs). As reported by the US department of transport, as of today, most traffic accidents are due to complicated road conditions and driver errors. Many studies have shown that adopting perfect autonomous vehicles (AVs) will remove human participation in driving, effectively reducing and avoiding traffic accidents [1]. Due to the rapid advancement of computer technology, artificial intelligence (AI), the internet of things (IoT), and sensors, autonomous driving technology have been evolved rapidly. With this rapid growth, there is no doubt that soon, intelligent vehicles will be the central pillar in driving safety, thus improving people's lives.

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In the last few decades, the defense advanced research projects agency (DARPA) launched the first project on autonomous land vehicles (ALV) [2]. Since then, AVs have gained significant interest from prominent research institutions and major companies like Google, which announced the launch of a research team to work on autonomous driving technology in 2009, Baidu, which launched the NonManCar project in 2013, and many other automobile manufacturers. Despite extraordinary efforts from these leading names in AV research, AVs still need to be out of reach except in limited trial programs due to critical concerns about their reliability and safety. In 2014, the Society of Automotive Engineers (SAE) published the autonomous driving standards, classified from level 0 to level 5 [3]. It is expected that L4 and L5 will revolutionize transportation by completely removing human participation when driving. Currently, even the most advanced AVs are still conditional automation at Level 3 [3], i.e., allowing driver “eyes-off” most of the time but with occasional intervention. One major challenge is the accidental failure of system perception, which reduces safety factors and even causes fatalities.

Generally, the IV system has three main components: environment perception, behavioural decision-making, motion planning and control. As mentioned above, environment perception is the main challenge for IVs, making it a major component for them, and the performance of environment perception directly stimulates the quality of vehicle decision-making and control. Therefore, an accurate environment perception system is necessary to reduce traffic accidents [4]. Target detection is so that IVs should have a perception ability to real driving situations. An IV uses sensors such as cameras, radar, and LiDAR to perceive its surrounding environment. Therefore, the present works focus on the sensor as the primary detection technology and summarize the vehicle detection methods in recent years. The paper walks through tasks, evaluation criteria, and existing public datasets of object detection for IVs and proposes vehicle detection's challenges and future development directions. The structure of this paper is shown in Fig. 1.

II. TASKS, EVALUATION CRITERIA, AND VEHICLE DETECTION DATASETS FOR INTELLIGENT VEHICLES

A. Tasks

For IVs to drive autonomously, they must provide pertinent information to the driver for safety, assistance, and comfort. Such information includes positioning and localization of

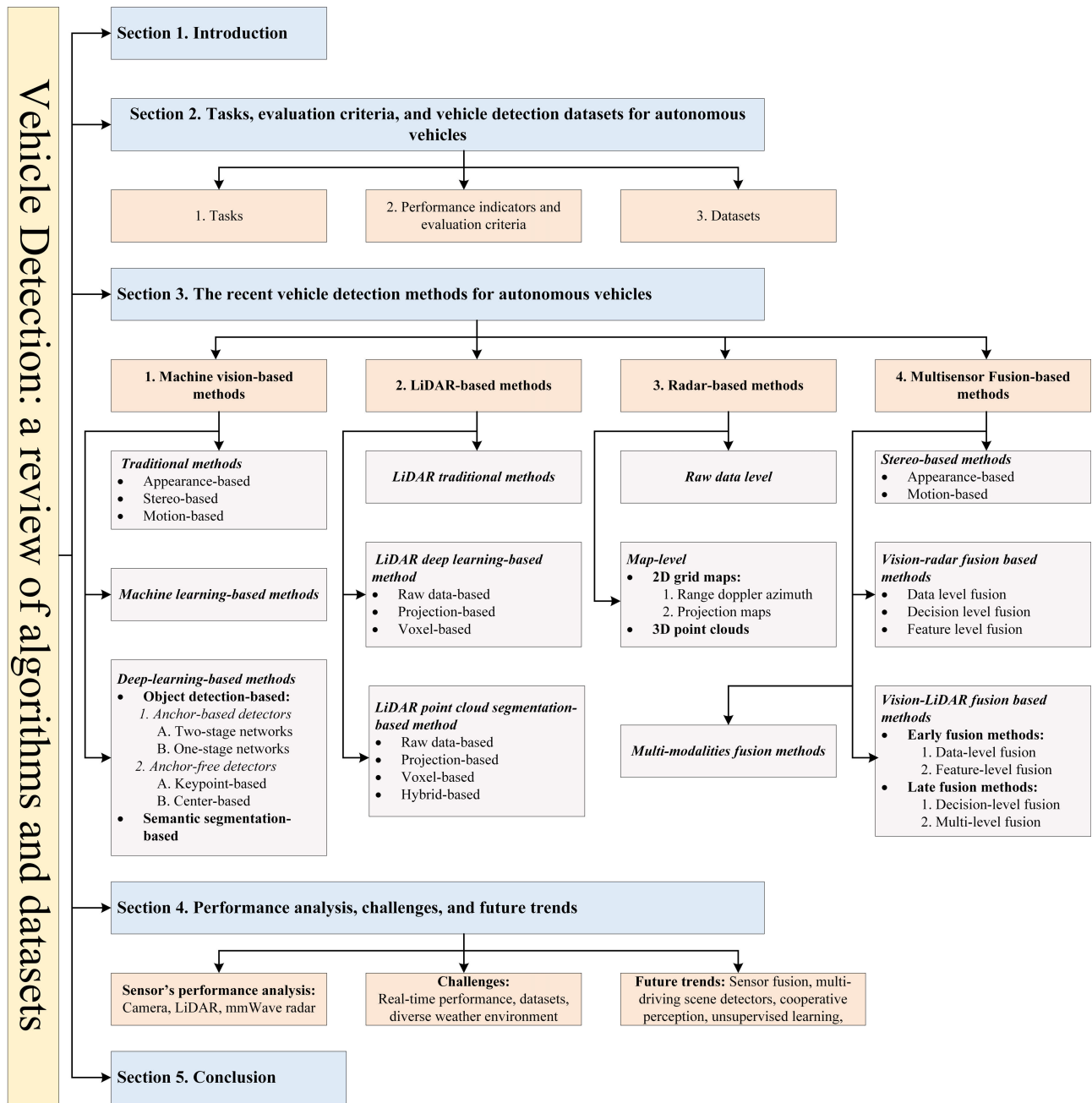


Fig. 1. The organization of this work.

obstacles or objects in the way of IVs. Vehicle detection is vital in this field since it involves classifying cars while detecting and positioning them in the input data. In the past decade, 2D object detection has become an essential feature of IV vision. 2D object detection methods mainly select the detected objects in the image vision of IV with a 2D bounding box, and then the selected objects are classified and positioned. However, detected objects are placed referring to the image rather than positioning them relative to the vehicle in the real world. 3D object detection has recently shown a significant impact over 2D object detection. Based on a 3D object detection mechanism, the system selects the detected objects in the vision image of the IV with a 3D bounding box, followed by the classification and positioning of these detected objects [5]. These systems not only position the selected objects in the

image but also determine where they are positioned in the real world.

B. Performance Indicators and Evaluation Criteria

Performance indicators are parameters used to assess the performance of the object detection model. Presently, average precision (AP) and average recall (AR) are used as the evaluation pillars for objection detection, where AP and AR represent accuracy and regression, respectively. The mean average precision mAP has to be calculated to represent the efficacy of the detection model, and by taking the average AP value of all classes, the mAP will be obtained. The precision-recall PR curve shows the trade-off between precision and recalls for different thresholds. It can be obtained by taking

the recall values as the horizontal axes and the precision values as the vertical axes.

Focusing on IVs, we take KITTI datasets as an illustration. For 2D object detection, the target positioning accuracy is obtained by collating whether the intersection over union (IoU) is greater than the threshold [6]. In object detection, the IoU is used to evaluate how close the prediction bounding box is to the ground truth on a particular dataset. For performance evaluation, the KITTI uses the mAP model from Pascal VOC [7]. For calculation, set 11 equally spaced thresholds in a range of [0,1]. For any recall value higher than each threshold, there is a maximum precision value, and to get mAP, you have to find the average of those 11 precision values.

Presently, 3D object detection is frequently used in IVs. The KITTI officially specified that for cars, correct prediction requires that the predicted 3D box overlaps the real 3D box by at least 70%, whereas for pedestrians and bicycles, at least 50% is needed [8]. Moreover, KITTI embraced the suggestions of the Mapillary team. They suggest that 40 recall positions could calculate the mAP more accurately than the 11 recall positions [9].

C. Datasets

The availability of high-quality annotated data accelerates the field of autonomous driving and machine learning. To this end, many companies and institutions released a bunch of annotated datasets that are believed to stimulate startups and academic researchers working on autonomous driving. Some of the most known datasets for vehicle detection include KITTI datasets [6], nuScene [10], Waymo open datasets [11], ApolloScape [12], CityScape [13], Oxford radar robot car [14], H3D [15], Lyft level 5 [16], Pandasets [17], ONCE (one million scenes) [18], AIO Drive [19], Ford multi-AV seasonal dataset [20], DDAD [21], CADc (canadian adverse driving conditions) [22], A2D2 [23], A*3D dataset [24], Argoverse [25], BLVD (building a large-scale 5D semantics benchmark for autonomous driving) [26], Rope3D [27], Itasca365 [28]. The description and the features of different datasets are summarized in Table I.

III. THE RECENT VEHICLE DETECTION METHODS FOR AUTONOMOUS VEHICLES

A. Machine Vision-Based Vehicle Detection

In the past decades, machine vision system has become a fascinating and promising area of research with a tremendous direct impact on different detection applications. Most studies have proven that detecting and recognizing vehicles plays a vital role in IVs. Vision based-sensors, often defined as passive sensors, have the intrinsic advantage of obtaining rich perception information (like lane markings, obstacle identification, texture, color, and semantics) from the traffic environment without altering it. Such vital information contributes to expressing the details of vehicle targets.

Initially, sensor devices like infrared cameras, sonars, and radars, were derived from expensive military applications. However, due to the interest in these applications and the

growing power of computers, today's technology offers these sensors at very competitive prices attracting big companies like Tesla [29] to choose different types of cameras as the essential solution for their IVs environment perception system. A fundamental task of FCAS (forward collision-avoiding system) is vehicle detection, which strongly affects the IV system's reliability.

According to the principles of the existing algorithm, machine vision-based vehicle detection can be categorized as traditional-based, machine-learning-based, and deep-learning-based methods.

1) *Traditional Methods for Vehicle Detection*: Two main stages exist in traditional vehicle detection methods. The first stage is the hypotheses generation (HG), where a brief analysis identifies the region of interest (ROI) throughout the image based on vehicle features. And the second stage, hypothesis verification (HV), confirms whether the ROI candidates contain vehicle targets. Therefore, the HG is the backbone of the HV since it serves as the input for the HV. The HG methods are often classified as appearance-based, stereo-based, and motion-based.

a) *Appearance-based methods*: Compared to the traffic environment, on-road vehicles have intrinsic appearance features that can exploit standard vehicle features in different traffic scenes. These features include edges, symmetry, texture, color, vehicle lights, and shadows, as shown in Fig. 2.

Symmetry-Based Methods: Most cars' rears are symmetrical. The symmetry-based methods utilize the symmetry to a vertical center line of the car rear [30] and searching regions set on the ROI containing vehicles in the image with high horizontal symmetry. Symmetry can also be used to adjust and optimize the bounding box of cars which are then used in the HV stages to verify whether the ROI contains vehicles [31]. However, the symmetry computation results in time consumption costs, making them inefficient in real-time applications. Moreover, brighter regions on the vehicle due to illumination change may result in loss of symmetry, causing loss of true positives.

Edge Feature-Based Methods: The rear view of a car can provide rich information for vehicle detection, the vehicle contours, bumper, rear window, license plates, etc. And these features have a robust linear texture in the vertical and horizontal directions [32]. Therefore, analyzing and grouping vertical and horizontal edges in the image has been used to determine the possible position of the car and its bounding box [33], [34]. However, these edge features information is susceptible to the texture of the image background since, most of the time, the background object may merge with the car contour or even present regions with edge density similar to the rear car, thus generating false positives.

Color-Based Methods: By applying different color channels and segmenting the thresholds of the vehicle color from the image background, you can separate the car from the environment. However, the color feature-based methods are sensitive to illumination changes and specular reflections, resulting in the loss of true positives [35], [36].

Taillight-Based Methods: Vehicle detection in the nighttime is based on the taillight. Generally, the taillights are red, making them easy to extract with a morphological filter

TABLE I

SUMMARIZED FEATURES OF DIFFERENT DATASETS SC. STANDS FOR SCENES, CL. FOR CLASSES, AN. FOR ANNOTATIONS, 3DB. FOR 3D BOUNDING BOXES, AND SE. FOR SENSORS. WHEREAS CL STANDS FOR CAMERA AND LiDAR, AND CLR FOR CAMERA, LiDAR AND RADAR SENSOR SUITE

Dataset	Sc.	CL.	An.	3DB.	Se.	Featured applications	Institution	Place
KITTI [6]	22	8	15k	200k	CL	Tasks of optical flow, visual odometry, 3D object detection, and 3D tracking for autonomous driving.	Karlsruhe institute of technology & Toyota American Institute of technology.	Karlsruhe Germany
nuScene [10]	1k	23	40k	1.4M	CL	It was taken in dense traffic and highly challenging driving situations encouraging researchers to develop methods that enable safe driving in urban areas.	Motional	Boston & Singapore
Waymo open [11]	1k	23	4	12M	CL	With a wide variety of environments, objects, and weather conditions (downtown, suburban, day, night, pedestrians, cyclists, construction, diverse weather)	Waymo	Different cities
Apollo Scape [12]	-	35	130k	70k	CL	It contains much more extensive and richer labels, including holistic semantic dense point cloud for each site, lane mark labels, 3D car instance, high accurate location for every frame in various driving videos from multiple sites, cities, and daytimes	Baidu Inc.	10 cities in China
City Scape [13]	-	30	30k	-	CL	Perception and tracking for autonomous driving	Daimler, Max planck institute informatik	50 cities
Oxford RobotCar [14]	-	-	-	-	CLR	The sensor suite a radar robust to environmental conditions such as fog, rain, snow, or lens flare, which typically challenge other sensor modalities.	Oxford robotics institute	Oxford UK
H3D [15]	160	8	27k	1.07M	L	To stimulate research on full-surround 3D multi-object detection and tracking.	Honda	San Francisco Bay
Lyft Level 5 [16]	-	-	-	-	CL	Used for motion prediction with over 1000 hours of data.	ride-sharing company Lyft	Palo Alto, California
Panda Sets [17]	8	28	80k	-	CL	Perception and tracking for autonomous driving.	Hesai & Scale	-
ONCE [18]	1M	5	15k	-	CL	3D object detection for autonomous driving.	Huawei Corp	China
AIO Drive [19]	-	-	-	10M	CLR	Perception for autonomous driving. The datasets have high-density, long-range point clouds.	Robotic institute, Carnegie Mellon university	-
Ford Multi-AV [20]	-	-	-	-	CL	Seasonal variations in weather, lighting, construction, and traffic conditions are experienced in dynamic urban environments.	Ford AV LLC	Michigan USA
DDAD [21]	-	-	-	-	CL	Used for long-range(up to 250m) and dense depth estimation in challenging and diverse urban conditions.	Toyota research institute	5 cities /USA 2 cities/ Japan
CADC [22]	75	10	7k	-	CL	Perception for autonomous driving in adverse weather.	University of Waterloo	Waterloo Canada
A2D2 [23]	-	-	12k	-	CL	Perception for autonomous Driving.	AUDI	3cities/ Germany
A*3D [24]	-	7	230k	-	CL	Datasets with a significant diversity of the scene, time, and weather used for the environment perception of autonomous driving.	-	Singapore
Argo-verse [25]	1k	-	-	-	CL	Tasks of perception for autonomous driving	Carnegie Mellon Mello university Georgia institute of technology.	Pittsburg & Miami USA
BLVD [26]	654	3	249k	-	CL	A large-scale 5D semantics benchmark aims to provide a platform for dynamic 4D (3D+interactive) interactive event recognition, & intention prediction.	Xi-an jiaotong university	Changshu China
Rope3D [27]	26	12	-	1.5M	CL	The roadside perception 3D datasets with the aim to support cooperative perception for AD.	Institute for AI industry research, Tsinghua university.	-
Itacha 365 [28]	40	6	-	-	CL	The data were repeatedly taken on a 15 km route under diverse scenes weather,time, and traffic conditions	Cornell engineering	Ithaca, Ney York

representing the ROI of vehicles [37], [38], [39]. However, this method is only helpful for nighttime vehicle detection.

Underneath Shadow-Based Method: The on-road car has a stable underneath shadow with perfect light conditions.

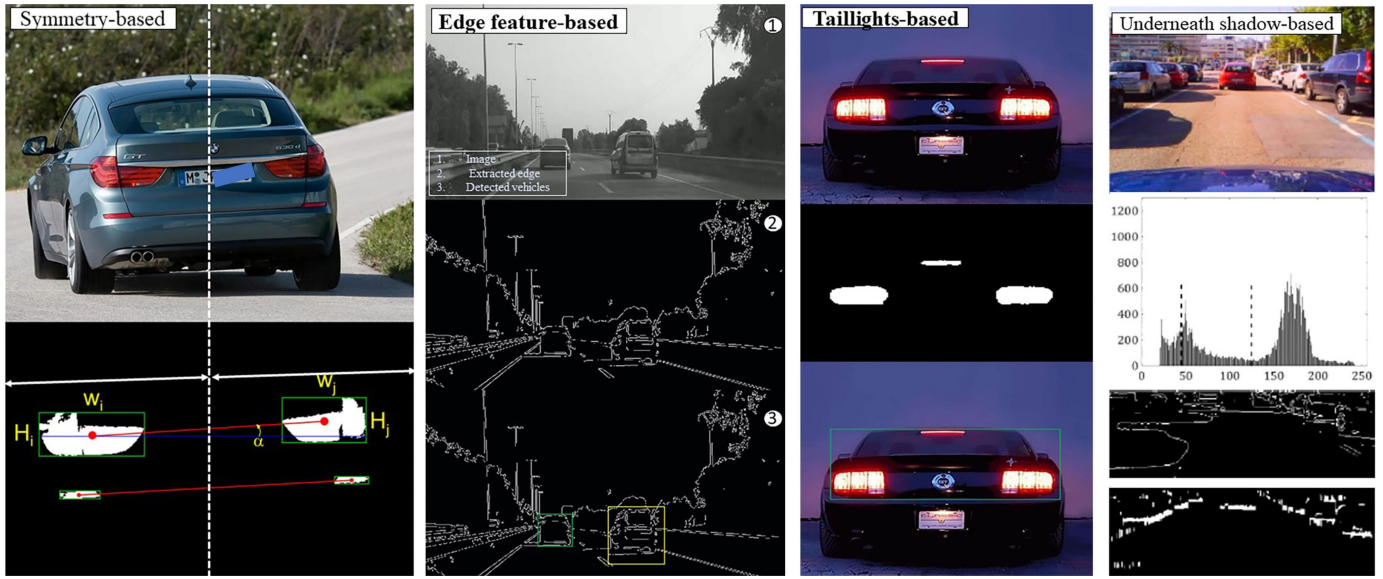


Fig. 2. Examples of feature and appearance based-methods.

Applying the segmentation thresholds can extract underneath shadow as the vehicle ROI in the HG stage [40], [41].

Texture-Based Methods: The road textures are homogeneous, whereas cars' surfaces are not, since they contain regions with high intensity variations. The texture-based method then relies on this difference to detect a vehicle from its surrounding environment [42]. However, it gets complicated when the image background contains other objects with texture as of vehicle, thus making it inefficient, especially in the urban environment.

The appearance-based methods are fast, convenient, and work under simple principles at a low computational cost. However, they are inefficient in real-world scenes since the vehicle ROI may contain noises from their surrounding environment and cannot efficiently represent all helpful information. Studies have shown that combining different feature methods can improve the performance of car detection [34]. Moreover, the ROI extracted by different appearance features techniques can be fed to machine learning (ML) or deep learning (DL) models to detect vehicle areas saving computational cost while improving vehicle detection [38], [41].

b) Stereo-vision-based methods: These methods mostly use two or more cameras to exploit the correlation between the two images of the same scene from the two cameras, which is a time-consuming task [43].

c) Motion-based methods: On-road vehicles are typically in motion. Which can bring effects of ego car and relative motion. [44] involves tracking the motion of pixels between consecutive frames of an image sequence to obtain the optical flow of the vehicle. However, this requires processing many frames to detect the vehicle, resulting in a high computational cost.

2) Machine Learning-Based Methods for Vehicle Detection:

With the intelligent transport system (ITS) development, the traditional method seems to lag behind the race. Due to the expeditious growth of computer technologies, ML became a great interest topic in vehicle detection. The ML detects

vehicles by using artificially designed features from the encoded image information of the cars. The high dimensional image space data are mapped to low dimensional space data to efficiently train the ML model, which is then used for car detection. Generally, in the ML for vehicle detection, the image processing to extract ROI is followed by a feature extraction fed into a classifier to be trained and optimized to detect non-vehicle and vehicle targets, as shown in Fig. 3.

The two main tasks for any ML model for vehicle detection are feature extraction and classifier training. It can be mentioned that local feature descriptors perform feature extraction. These are designed to extract and distinguish the vehicle's local features allowing a quick and efficient search of cars in the image. Ideally, excellent feature descriptors should remain stable to geometric and photometric transformation. Many studies have proposed several feature extraction methods, achieving incredible results in the vehicle detection methods, the most known ones include the histogram of gradients (HOG) and Harr-like features.

a) HOG: Originally designed for pedestrian detection [45], then expanded to vehicle detection. Some studies proved that combining HOGs can extract excellent vehicle features compared to the original HOG. Some examples include two HOG vectors [46], the pyramid of histogram of gradient (PHOG) [47], three HOG configurations (horizontal, vertical and concentric HOG) [48], and symmetry HOG [49].

b) Haar-like feature: Viola and Jones [50] proposed this feature descriptor for human face recognition, and then it was expanded into vehicle detection since it is simple and easy to compute [51]. However, the haar-like feature vector generated from the image is high [52], which needs to process further operations to reduce the computation cost or hardware storage. Despite the achievement of those mentioned methods, other feature extraction methods also showed remarkable achievements in vehicle detection. These include Gabor features [53], speed up robust features (SURF) [54], etc.

After generating the local features from image sequences, the extracted features have to be identified by an ML classifier.

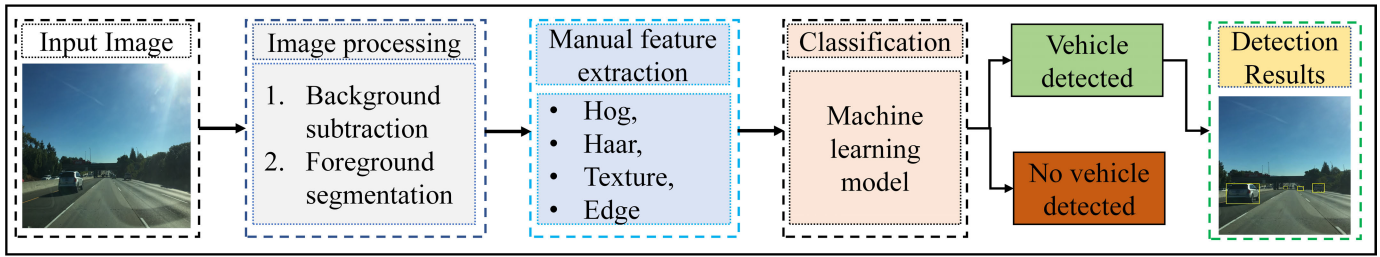


Fig. 3. General process of machine learning based vehicle detection.

TABLE II
SOME SELECTED WORKS ON FEATURE-CLASSIFIER COMBINATION

Features	Field of View	Classifier
Gabor	Rear-view	Backpropagation model [53]
Haar-like	Side-view	Adaboost [51], [55]
	Rear-view	Neural network [56]
	Rear-view	KNN [57]
HOG	Rear-view	Adaboost [46]
	Rear-view	SVM [49], [55]
PHOG	Front/Rear-view	GA-SVM [47]
SURF	Rear-view	SVM [58]
SIFT	Rear-view	Adaboost [59]

Generally, two class patterns known as vehicle and non-vehicle targets are classified [51]. The method involves additional work, such as collecting the positive sample (vehicle targets) and negative samples (non-vehicle targets) from different sources and training them with a classifier. The classifier learns the behavior of vehicle appearance statistically and draws a boundary between positive and negative samples.

K-nearest neighbour (KNN), support vector machine (SVM), AdaBoost, and decision tree are the representative classifiers in ML vehicle detection. The choice of a classifier depends on the balance between generalization potential, fitting accuracy, and the dimensional data of the extracted features. Note that a substantial within-class variability should be extracted to achieve a good classification performance. Recently, more studies have shown that combining two or more feature classifiers can generate fantastic vehicle detection results. Selected works for combining feature extraction and classifiers are shown in Table II.

The ML-based method achieved good results in vehicle detection. However, their classifiers scan the entire image, increasing the computational cost. It is shown in [60] that over half of the image area contains no vehicle information. To deal with this issue, we can use a combination of the traditional methods with the ML classifiers to improve the real-time performance of a detector. Since the ML classifier will analyze an ROI from traditional methods instead of traversing the entire image to extract the vehicle feature. Perfect illustrations can be found in [46].

3) Deep-Learning-Based Methods for Vehicle Detection:

Decades ago, the essence of object detection was to locate and classify objects using 2D rectangular boxes around detected objects in the image. Due to the limitation of the computing power and datasets, training a high-performing DL model with no overfitting was cumbersome, which limited the application of DL models for object detection a few decades ago [61]. Therefore, traditional object detection algorithms

were still popular. With the advance in computation resources, DL achieved a remarkable development. At this time, the CNN-based network achieved unrivaled success in recognition tasks [62], which led to the rapid growth of CNN-based object detection (OD) models. Some of the most pioneering CNN-based OD models include RCNN [63], Fast RCNN [64], Faster RCNN [62], Yolo [65], [66], [67], [68], [69], and SSD [70], to name a few.

Recently, many studies have shown that object detection has some relationship with object classification, semantic segmentation, and instance segmentation. Object classification only categorizes or identifies objects in the image. In contrast, object detection not only identifies the category of objects but also locates the objects with rectangular 2D bounding boxes in the image. Semantic segmentation, however, only needs to predict the categories of each pixel, while instance segmentation predicts each pixel's categories and needs to the object instances. Fig 4 illustrates these relations. It can be mentioned that most OD models can be applied in vehicle detection scenes. Currently, relevant studies show that vehicle detection networks can be categorized into object detection-based and segmentation-based networks.

a) Object detection-based networks: Generally, object detection network models can be categorized into anchor-based and anchor-free models. In anchor-based models, regressing object localization is based on the pre-defined anchors of the original image or feature maps. Unlike the anchor-based model, the anchor-free model does not need to design and preset anchor boxes to fit the target; instead, it directly predicts an object's center point or key points, and groups them into one object to get bounding boxes.

Anchor-Based Detectors:

According to different training methods, anchor-based models can be divided into one-stage and two-stage networks. In two-stage networks, regional proposal boxes are generated to show the possible object targets, and a prediction network is then used to detect the object on these proposal boxes. In contrast, the one-stage-based methods return the object area on the feature map by giving the final prediction results.

Two-Stage Detection Frameworks: Two main vehicle detection tasks are performed in these frameworks, a car region proposal followed by car targets detection. The vehicle regional proposal boxes are generated showing the possible cars, and a prediction network is then used to detect vehicles on these proposal boxes. The representative frameworks include RCNN [64], Fast RCNN [71], Faster-RCNN [62], and SPP-Net [72]. In 2014, R. Girshick proposed an RCNN detection framework. The framework adopts the selective

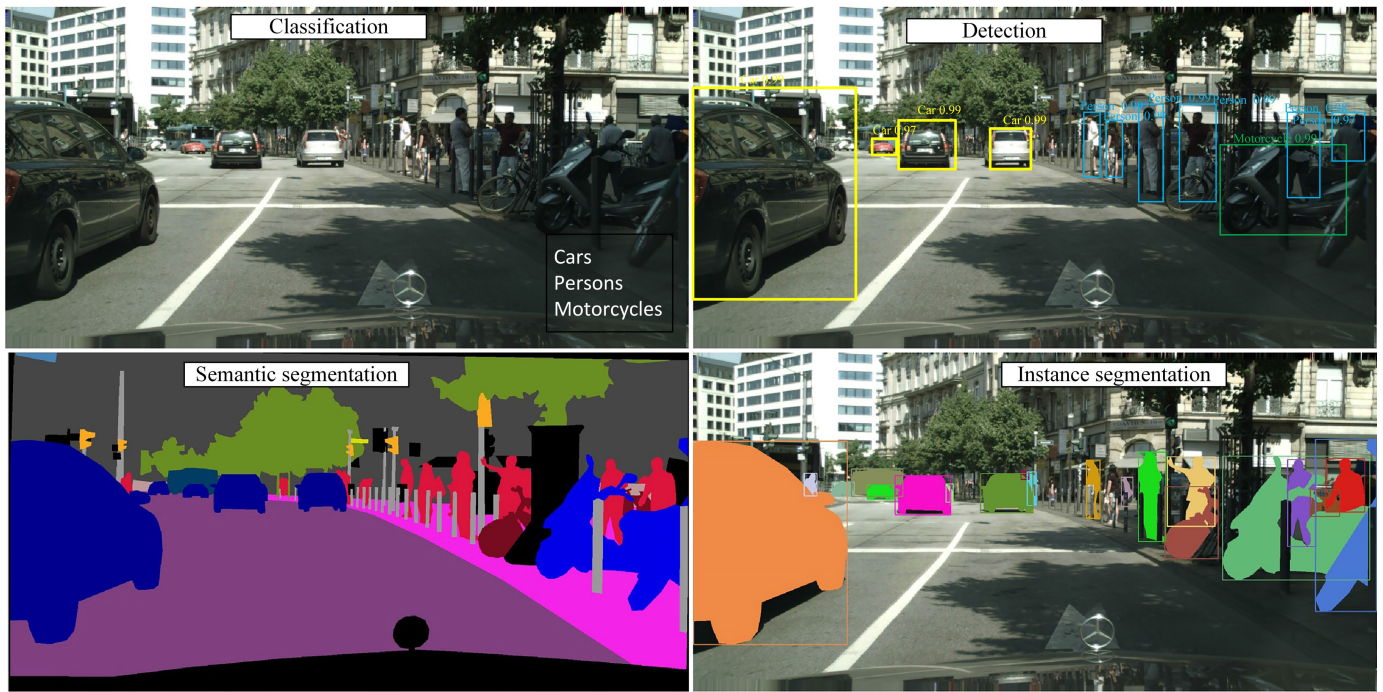


Fig. 4. Relationship between vehicle classification, detection, semantic segmentation, and instance segmentation.

search [73] to generate a region proposal box and then uses CNN to extract features. The SVM classifier and bounding box regression are trained and predict the final results. Even if CNN-based feature extraction improved the detection results, the frameworks also encountered too many problems like an arduous training process and long detection time, making it unable to be applied in the real world.

Fast RCNN [71] and Faster RCNN [62] were proposed to solve such issues. In Faster RCNN, the selective search algorithm was replaced by a regional proposal network (RPN), significantly reducing the time consumed by the regional recommendation.

One-Stage Detection Frameworks: These models remove the vehicle pre-selection box generation and unify vehicle identification and detection into one network or directly predict the vehicle centre and bounding box by setting a series of anchors on the feature map. Typical known frameworks include YOLO v1 [65], SSD [70], YOLO v2 [66], YOLO v3 [67], M2Det [74], YOLO v4 [68], and YOLO v7 [69]. YOLO was the first single-stage detector. The detector applies a single neural network to the entire image by dividing it into multiple regions. Then, the model predicts the bounding box and the probability of each region, resulting in high detection speed. Such high-speed models are needed in autonomous driving, especially for vehicle detection tasks. However, compared with two-stage vehicle detection algorithms, the YOLO accuracy is not good since it only detects a small number of vehicles from the anchor. Moreover, it gets even more inaccurate if small objects like pedestrians are detected.

YOLO v1 may be classified as an anchor-free model but suffers from a low recall rate. Therefore, a series of YOLO v2 and YOLO v3 as anchor-based models were proposed to optimize the YOLO v1's issues, which increased the accuracy and achieved the relative balance between speed

and accuracy. Wang et al. [69] developed an SSD one-stage network. This algorithm outputs feature layers of different sizes through multilayer mapping in the convolution layer to detect multiscale objects. Unlike YOLO, the SSD detector increased the detection accuracy of small entities. In 2020, YOLO v4 [68] was proposed. The network adopted the CSPNet structure [75] as the backbone and used SSP [72] and PAN structure [76] to perform the feature fusion in the neck part. Moreover, clustering algorithms were used to generate the predicted frame size making the networks which made the network achieve the best balance between the speed and accuracy of the detection. Currently, YOLO v7 [69] surpasses all previous state-of-the-art object detection models and other YOLO versions in both speed and accuracy. Hence, YOLO v7 is expected to become the industry standard for object detection in the near future.

Anchor-Free Detectors: The existing studies show that anchor-free models can be classified as key-point- and center-based. Key-point-based methods detect specific object points, such as center and corner points, and group them for bounding box prediction. The example models include CornerNet [77], CenterNet [78], ExtremeNet [79], RepPoints [80], and CornerNet-Lite [81]. In contrast, center-based methods directly predict the object's center point and perform the object bounding box regression. These make them more comparable to anchor-based detectors. Some center-based models include FSAF [82], FCOS [83], FoveaBox [84], FPN [85], and GA-RPN [86].

Generally, two-stage vehicle detectors are more accurate than one-stage vehicle detectors. In addition, they are good at detecting small vehicles due to their candidate region generations. Therefore, as of today, many vehicle detection algorithms rely on two-stage networks. However, the real-time performance required with the IVs makes one-stage networks

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT OBJECT
DETECTION NETWORKS WITH USED BACKBONE

Detector	Backbone	AP	AP50	AP75
<i>Anchor-based two-stage</i>				
FasterRCNN-fpn [85]	ResNet-101	36.2	59.1	39.0
FasterRCNN-grmi [87]	Incep Res v2	34.7	55.5	36.7
FasterRCNN-tdm [79]	Incep Res v2	36.8	57.7	39.2
MaskRCNN [88]	ResNet-101	38.2	60.3	41.7
FasterRCNN [62]	ResNet-101	34.9	55.9	37.4
Cascade RCNN [89]	ResNet-101	42.8	62.2	46.3
R-FCN [90]	ResNet-101	29.9	51.9	-
MLKP [91]	ResNet-101	28.6	52.4	31.6
SNIP [92]	DPN-98	45.7	67.3	51.1
<i>Anchor-based one-stage</i>				
YOLO V2 [66]	DarkNet 19	21.6	44.0	19.2
YOLO V3 [67]	DarkNet 53	33.0	57.9	34.4
YOLO V4 [68]	CSPDarkNet53	49.7	54.3	46.1
SSD [70]	VGG16	28.8	48.5	30.3
DSSD [93]	ResNet-101	33.2	53.3	35.2
RefineDet [94]	ResNet-101	36.4	57.5	39.5
RetinaNet [95]	ResNet-101	39.1	59.1	42.3
STDN [86]	DenseNet-169	31.8	51.0	33.6
<i>Anchor-free keypoint-based</i>				
ExtremeNet [79]	Hourglass-104	40.2	55.5	43.2
CornerNet [77]	Hourglass-104	33.0	57.9	34.4
CenterNet-HG [96]	Hourglass-104	42.1	61.1	45.9
Grid RCNN [97]	ResNet-101	43.2	63.0	46.6
CornerNet-Lite [81]	Hourglass-54	43.2	-	-
CenterNet [78]	Hourglass-104	44.9	62.4	48.1
RepPoints [80]	ResNet-101	45.0	66.1	49.0
<i>Anchor-free center-based</i>				
FCOS [83]	ResNet-101	43.2	62.8	46.6
Foveabox [84]	ResNet-101	42.1	61.9	45.2
FSAF [82]	ResNet-101	42.9	63.8	46.3
GA-RPN [86]	ResNet-50	39.8	59.2	43.5
ATSS [98]	ResNet-101	50.7	68.9	56.3

an inevitable future trend of vehicle detection technology. The comparison of anchor-based and anchor-free model's detection performances is shown in Table III and a brief explanation of some pioneering anchor-free and anchor-based models is shown in Table. S1 Whereas, the general architecture of one- and two-stage networks is shown in Fig. S1, S2 and their pros and cons are summarized in Fig. S3 of the supplementary materials.

In general, all the above models first use CNN to process an image in order to obtain a deep feature map. A backbone of the entire network is obtained by performing subsequent processing on this feature map. The detector's backbone can decisively influence the effect of the whole algorithm. The application of different detectors in real scenes relies on the model's compactness, detection accuracy, and speed. Therefore, choosing the backbone network for vehicle detection has to rely on those three things. Pioneering studies showed that the more layers in a backbone network, the stronger it is in extracting feature maps. Thus, the more accurate it is for vehicle detection.

As an illustration, Faster-RCNN will attain 62% of mAP and 17fps when trained with seven layers backbone (ZFNet), whereas the same model attained 73% mAP and 5fps when adopting a 16 layers backbone (VGG16) [62]. Therefore, it is crucial to balance the vehicle detection speed and accuracy to deploy your model in real scene. Table IV represents the performance comparison of some pioneering backbone

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT BACKBONE
NETWORKS. ACC: ACCURACY IN %

Backbone network	Params (Millions)	Top-1 acc	Top-5 acc	Year	ref
AlexNet	60	57.2	80.3	2012	[99]
ZFNet	58	64	85	2013	[100]
VGG-16	138	71.5	89.8	2015	[101]
GoogLeNet	6.8	69.8	93.30	2014	[102]
ResNet-50	23.4	79.3	96.4	2015	[103]
ResNet-101	42	80.1	96.4	2015	[103]
Inception-V2	12	79.9	95.2	2015	[104]
Inception-V3	23.5	82.7	96.5	2015	[105]
Inception-V4	42	83.5	96.9	2016	[106]

networks for object detection, whereas Table S2 of the supplementary materials summarize their novelty.

Deploying vehicle detectors in real time is cumbersome since you have to design an efficient, lightweight vehicle detector that can accurately and precisely extract vehicle features. Recently, relevant research has shown excellent results in developing such compact networks. These include SqueezeNet [107], Xception [108], MobileNet [109], ShuffleNet v1-v2 [110], [111], and PeleeNet [63]. The SqueezeNet [107] uses the fire module consisting of a squeeze layer and an expanded layer. The Xception [108] is an improvement over the InceptionV3 that replaces the convolution layers in the InceptionV3 with a depthwise separable convolution layer.

In the MobileNet [109], a convolution method different from the traditional convolution method known as the depthwise separable convolution is used. The ShuffleNet v1-v2 adopted point-wise group convolution instead of point-wise convolution [110], [111]. In the PeleeNet [63], two-way dense layers are used to obtain receptive fields of different scales, and feature learning of small-scale and large-scale objects is simultaneously performed. The performance comparison of various lightweight backbone networks is shown in Table S3 of the supplementary materials.

Balancing speed and accuracy is a long-standing issue for deploying deep learning models. The tradeoff is much more crucial for real-time safety-critical systems like autonomous vehicles. CNN has contributed to deep learning's success in computer vision over the last few decades. It is interesting to note that CNN's dominant position is challenged by the recently developed vision transformer (ViT) [112]. In contrast to CNNs, ViTs can achieve competitive performance on various visual benchmarks by applying the pure self-attention-based architecture to sequences of image patches. As a result, many scientists are now using transformers as the backbone of their vehicle detectors.

Scholars in [113] proposed a dynamic graph transformer 3D object detection network (DGT-Det3D) based on a dynamic graph transformer (DGT) module and a proposal-aware fusion (PAF) module. The DGT module is built on a dynamic graph and graph-aware self-attention module that adaptively concentrates on the foreground points and encodes the graph to capture long-range dependencies. Authors in [114] developed an effective approach for accelerating transformer-based 3D object detectors by dynamically halting tokens at different

layers depending on their contribution to the detection task. Researchers in [115] designed an important region-based reconstructed deformable self-attention (RDSA) module that shifts attention to important regions for efficient global modeling. And based on the RDSA, they proposed the swin deformable transformer backbone that improves the feature extraction ability and convergence speed.

VoxSeT [116] detects 3D objects from point clouds by means of set-to-set translation. The network was developed upon a voxel-based set attention (VSA) module, which reduces the self-attention in each voxel by two cross-attentions and models features in a hidden space induced by a group of latent codes. In VISTA [117], a plug-and-play fusion module, the multilayer perceptron is replaced with a convolution. CenterFormer [118] is a center-based transformer network for 3D object detection. The model uses a center heatmap to select center candidates on top of a standard voxel-based point cloud encoder, and then adopts the feature of the center candidate as the query embedding in the transformer.

SWFormer [119] is the sparse window transformer for 3D object detection. The model transforms 3D points into sparse voxels and windows, then efficiently processes these variable-length sparse windows using a bucketing scheme. STVD [120] was developed to efficiently solve the multi-scale feature extraction problem. The model uses hierarchical feature maps and complete information flow between picture patches to solve that issue. VoTr [121], known as voxel transformer, works as a novel and powerful voxel-based transformer backbone for 3D object detection from point clouds. MFT-SSD [122], known as multi-source features fusion single stage 3D object detection with transformer, combines point-based and voxel-based backbones, and extracts point, multi-scale voxel and BEV features as multi-source features. With the recent success of transformer in visual recognition, there is no doubt that vision transformer is an expected hotspot in computer vision applications. It can be mentioned that these transformer-based detectors can also be applied for the vehicle detection.

b) Semantic segmentation-based detection networks:

Semantic segmentation associates a label or a category with every pixel in the image. Compared to target level-based detection methods, semantic segmentation-based methods seem more accurate and precise since they recognize a collection of pixels that form distinct categories. Moreover, they are good at expressing the objects' location and contour, which is highly important for AV environment perceptions [13]. It can be mentioned that all semantic segmentation methods are applicable in the field of vehicle detection. Due to safety priorities in autonomous driving, only fully supervised semantic segmentation methods are being adopted. These methods can be classified as regional classification-based and pixel classification-based methods. The different state-of-the-art models can be found in Fig. S4 of the supplementary materials.

The regional classification method can be compared to a two-stage network where an RPN extracts car candidate regions while a trained classifier assigns every pixel in the image to a label. Some of the most known models include Deep Mask, a vehicle proposal algorithm based on

CNN. The model generates a class-agnostic mask and an associate score which estimates the likelihood of the patch fully containing a centred car [102]. Sharp Mask is an extension of deepMask that generates higher-fidelity masks using an additional top-down refinement step [123]. Mask RCNN uses anchor boxes to detect vehicles of different scales and overlapping cars in an image [88]. Even if these models are likely to be more accurate, the dependence on candidate regions generation limits the real-time deployment of autonomous driving. Thus, massive work is still needed to improve real-scene performance.

Upon this issue, some pixel classification-based models were proposed to solve these problems. In 2015, a fully convolutional network (FCN) for semantic segmentation was proposed [124]. The network employs locally connected layers, such as convolution, pooling, and upsampling, without using any dense layer. To improve the FCN, the DeepLab [125] by Google Lab was proposed. Here, the dense prediction is achieved by simply up-sampling the output of the last convolution layer and computing pixel-wise loss. DeepLabv2 [123] is a semantic segmentation architecture built on Deeplab with an atrous spatial pyramid pooling scheme ASPP [125]. Deeplabv3 [126] is an architecture that improves upon deeplabv2 where segmenting objects at multiple scales; model modules are designed to employ atrous convolution in cascade or parallel to capture multiscale context by adopting considerable atrous rate. Studies have shown that the DeepLab series effectively increase the filter's receptive field, improving the precision and accuracy of a semantic vehicle detector. However, deploying Deeplab series models in real-life IV scenes is computationally expensive.

To leverage the computation cost with speed and accuracy, feature fusion models have been proposed. The model integrate related information extracted from a group of training and testing vehicle images without losing any data [127]. The cross-layer fusion of features dramatically reduces the computational power and memory consumption caused by probabilistic graph computation. Some state-of-the-art examples include PSPNet, known as the pyramid scene parsing network, which utilizes a pyramid parsing module that exploits global context information through different region-based context aggregation [128]. ICNet uses deep supervision and runs the input vehicle image at different scales, each scale through their subnetwork and progressively combining the results [129].

RefineNet [130] proposes an efficient and generic way to fuse coarse high-level features with finer-grained low-level features to generate high-resolution semantic features. Some scholars tried to adopt generative adversarial networks (GANs) to improve vehicle semantic segmentation detection models. However, the GAN-based model training and optimization is still cumbersome since the model easily collapses, does not converge, and is unstable. Such models include those given in [131]. The performance detection results of the different complex and lightweight models on the cityscape dataset [13] are shown in Table V, and their segmentation results are shown in Fig. S5 of the supplementary materials.

The computational complexity of the semantic segmentation vehicle detection models is a current burden in autonomous

TABLE V
THE PERFORMANCE COMPARISON OF STATE-OF-THE-ART
SEMANTIC SEGMENTATION MODELS

Model	Input (PX)	Params (M)	mIoU (%)	Frame (fps)	Year	Ref
DeepLab	512*1024	262	63.1	0.25	2017	[125]
SegNet	640*360	29.5	57	16.7	2017	[132]
PSPNet	713*713	250.8	71.2	0.78	2017	[128]
ICNet	1024*2048	26.5	69.5	30.3	2018	[129]
BiSeNet2	768*1536	49	74.7	45.7	2018	[133]
DFANet A	1024*1024	7.8	71.3	100	2019	[134]
DFANet B	1024*1024	4.8	67.1	120	2019	[134]
ENet	640*360	0.4	57	135.4	2016	[135]
LEDNet	1024*512	0.94	70.6	71	2019	[136]
ESPNet	515*512	0.4	60.3	112	2018	[137]
CGNet	680*680	0.5	64.3	50	2020	[138]

driving [139]. These models' execution time and memory usage are not good compared to other car detection techniques. Therefore, as future works, scholars should focus on designing lightweight models that meet the accuracy and speed requirements and are affordable to be deployed in our daily use devices [140].

Recently, lightweight vehicle semantic segmentation models have been developed, and their studies showed promising and excellent results. These models include DFANet, published in 2019 as deep feature aggregation for real-time segmentation, starting from a single lightweight backbone and aggregate discriminative features through sub-network and sub-stage cascade, respectively [134]. The experimental results show the superiority by proving that DFANet is eight times lesser FLOPs and two times faster than the existing real-time semantic segmentation techniques. The model achieved 0.703mean IoU on the CityScape dataset with 1.7GFLOPs at a speed of 160fps on only one NVIDIA Titan X GPU.

LEDNet is a lightweight network employing an asymmetric encoder-decoder architecture for real-time semantic segmentation. The experimental results confirm the compactness of the model while its accuracy is still adequate [136]. The Models parameters are less than 1M, and it achieved 71 fps, and 0.706 mean IoU on a single GTX 1080ti GPU. Some failures of an image processing-based deep learning model are due to either insufficient training data or a cumbersome process of labelling data. To tackle such issues, you either improve the model or make the datasets more diverse.

B. LiDAR-Based Methods for Vehicle Detection

LiDAR is a sensor that perceives the environment by transmitting laser signals and receiving the reflected signals from the obstacles in the environment. The perception information is in the form of point clouds containing the objects detected, their size, and their exact position. These active sensors can accurately obtain 3D geometry information of vehicle surrounding [141]. Moreover, they are not sensitive to light conditions making them more applicable in AV fields. However, they come with a computational burden since their data are sparse and highly unstructured. Practically, existing LiDAR-based methods can be classified as traditional and deep learning techniques.

1) *Traditional Methods*: Most traditional methods apply some techniques to pre-process data followed by ground removal. Then cluster point cloud is followed by feature extraction, which is used for vehicle classification. Due to the sparsity of the LiDAR point cloud data, most researchers pre-process the data to reduce the computational burden. These techniques include an occupancy map [142], a plot method [143], and a range image [144]; These techniques lose plenty of information since they transform 3D LiDAR data into a 2D or 2.5D shape. It has been shown that removing some points will increase the real-time performance of vehicle detectors. To cope with this issue, scholars [145], [146] proposed a method to set up a feature threshold to remove road point clouds since they differ from other obstacles in a traffic environment, reducing the computation burden. The process of thresholding road points cloud is shown in Fig. 5.

However, the method fails when road pavement fluctuates or presents many potholes or slopes. To solve this issue, an assumption of making the uneven road an assembly of many planes was derived. And by applying fitting algorithms like MRF (Markov random sample) [147], GPR (gaussian process regression) [148], and RANSAC (random sample consensus) [149], the fluctuated road can be assembled as a plane road. After removing unusable data points, the valuable data points have to be clustered. Most scholars adopt K-Mean clustering or the DBSCAN clustering algorithm [150] extracted feature maps are then used to detect the vehicles, which is mainly performed with ML algorithms such as decision trees [151], random forest [152], or KNN [153]. These methods are simple to implement and reliable in preprocessing data. However, their classification and segmentation rules ignore massive point cloud information, making their precision unrealistic. Moreover, the multi-steps-based detection process makes these methods unsuitable for real-scene applications.

2) *LiDAR Deep-Learning-Based Method*: Recently, with the increase and development in computational power, deep learning achieved a significant performance over traditional techniques for vehicle detection in terms of accuracy and detection speed. Deep learning embodies the capacity of complex feature representation generalized to point cloud point data [154]. Moreover, LiDAR deep learning-based vehicle detection models are structured into end-to-end learning, making them applicable to real scenes. Considering the data type where we extract feature maps, LiDAR deep learning methods can be categorized as data-based, projection-based, and voxel-based, as shown in Fig. 6.

a) *Raw data-based methods*: These methods apply 3D object detection techniques directly from raw point cloud data. Some scholars [155] adopt 3D CNN to extract vehicle features in the raw point cloud, which significantly preserves the original information of the LiDAR. PointRCNN [156] did the 3D object detection directly from raw point cloud data. Their framework adopts two stages. The first stage is for bottom-up 3D proposal generation, and the second stage is for refining proposals in canonical coordinates to obtain the final detection results. However, the irregularity of raw point clouds made these methods not robust enough to meet the safety level

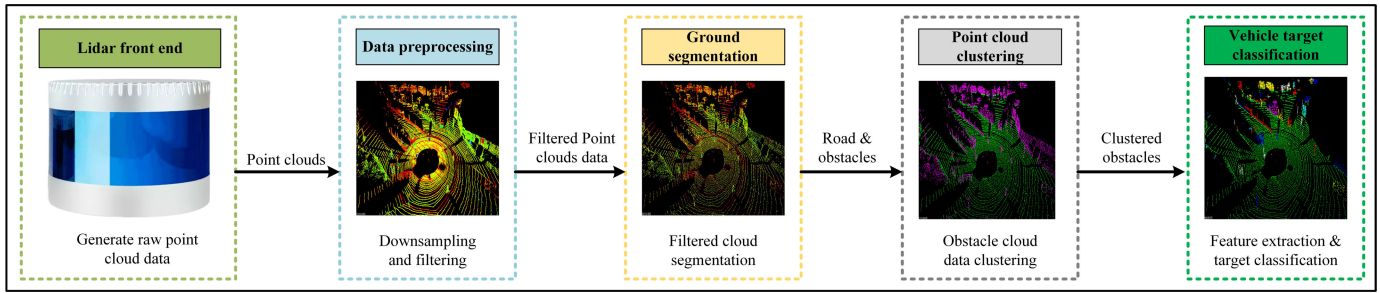


Fig. 5. Traditional algorithm process for vehicle detection using LiDAR point cloud.

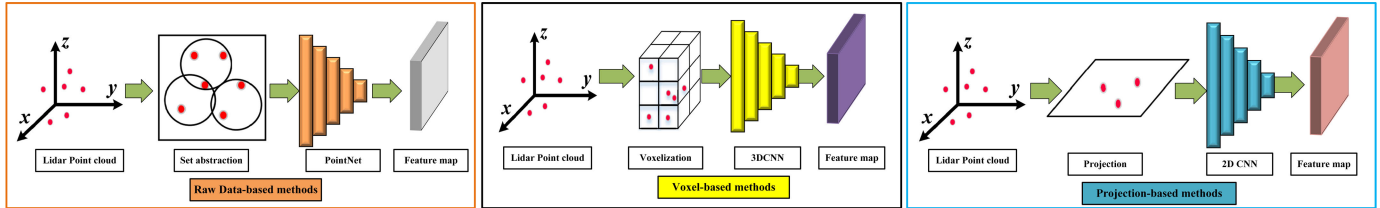


Fig. 6. Different categories of LiDAR deep-learning-based vehicle detection techniques.

in vehicle detection. Point cloud data have some advantages. First, point clouds are the point; therefore, they are invariant to the permutations of their members. In addition, they can preserve geospatial information and their local structure. Some researchers exploited the uniqueness of raw points to cope with the issues mentioned above.

PointNet [157] presents deep learning for reasoning about 3D geometric data such as point clouds or meshes. The model designed a novel neural network that directly consumes point clouds and respects the permutation invariance of points in the input. And this solved the problem of unordered sets from a data structure. However, independently dealing with each sampling point only considers global vehicle features, which limits the model's vehicle detection performance. PointNet++ [158] finds both global and local vehicle features, solving unordered sets data structure and improving the model's generalization.

LiDAR raw data-based models achieved remarkable success in vehicle detection and contributed much to the autonomous driving field. However, treating points at local scales make the models learn the neighborhood features point in a separate form leading to failure in preserving spatial information between neighboring local points.

b) Projection-based methods: These methods project 3D point clouds to 2D maps to benefit from existing well-developed vehicle detection algorithms. The projection is either front view (FV) Or bird's eye view (BEV). Authors in [154] rely on FCN but implant it in 3D convolution operation, generalizing vehicle detection. The model transforms the LiDAR point into red-green blue-depth (RGBD) camera images, predicting objectiveness and object shape in 3D Space.

Vote3Deep operates like this model; however, it uses a sliding window strategy for final prediction, which is unsuitable for varying sizes and directions. DepthCN [159] proposed a vehicle detection based on HG and HV where the data from 3D LiDAR were transformed into depth maps. The authors start by removing ground points, followed by point cloud segmentation. Segmented vehicles are projected to the

depth map, and bounding boxes are fitted to the car targets. Bounding boxes are then sent to CNN for classification as the vehicle or not.

SqueezeSeg [160] projected the LiDAR point cloud into FV, and a CNN followed by a conditional random field (CRF) was adopted. A DBSCAN [150] is used to cluster 3D points. The authors in [161] proposed a vehicle detection method using CNN on 2D image data. The module project 3D bounding boxes in the two-dimensional camera space. RT3D [162], a 3D LiDAR point cloud is projected onto a depth map bird's eye view encoded with height information of points, and then the RPN classifies the proposed regions with pre-ROI-pooling. BirdNet [163] projects LiDAR point into BEV and adopts CNN to track the position and posture of the vehicle and calculate the height to obtain the 3D detection information of the car targets. The authors in [56] adopted YOLOv3 for vehicle detection in BEV and used deepGAN (cycle GAN) to detect the vehicles.

Projection-based methods benefit from a well-developed image-based 2D detection algorithm, improving their performance compared to raw data-based models. However, projecting a 3D point cloud to a 2D map inevitably loses or distorts useful 3D spatial information reducing these models' detection accuracy.

c) Voxel-based methods: Non-uniformity sampling of 3D space, the effective range of sensors, occlusion, and relative pose are some factors that make the LiDAR point cloud sparse and have highly variable point density. To solve this problem, point clouds are divided into regular voxel grids of a specific size to describe the distribution in 3D space. The representative models include VoxelNet [164], an end-to-end model that simultaneously learns a discriminative feature representation from point clouds and predicts accurate 3D bounding boxes. The model works on sparse 3D points and efficient parallel processing on a voxel grid by avoiding the information bottlenecks by manual feature engineering.

VoxelRCNN [165] takes voxels as input, generates dense regional proposals from bird-eye-view feature representation, and adopts voxel RoI to extract feature region features from

a 3D voxel for further refinement. PV-RCNN [166] integrate multiscale 3D voxel CNN and PointNet-based features to a small set of critical points by the new proposed voxel set abstraction layer. The ROI-grid points are then used to extract much richer context information for the proposal refinement. Other voxel-based methods exist, such as 3D-ShapeNet [167] and SECOND [168].

The voxel-based methods achieve a realistic balance between accuracy and efficiency by fully exploiting voxel representation. However, transforming the raw data point cloud to a voxel grid inevitably causes the model to lose some information, reducing performance. Table VI summarizes the performance of different LiDAR deep learning state-of-art models on the KITTI test datasets for car detection. And Fig. S6 of the supplementary materials represents some of the qualitative results for these models.

3) *LiDAR Point Cloud Segmentation-Based Methods*: The point cloud segmentation methods assign per-point semantic labels helping in the scene understanding. The semantic class labels include different elements of the road scenario, like vehicles, buildings, pedestrians, road curbs, road pavement, and trees. The existing point cloud segmentation-based methods typically fall into two main categories. There are traditional-based and deep-learning-based methods. Traditional semantic segmentation methods rely on complicated hand-crafted features such as surface normal or generated descriptors and hard threshold decision rules based on some clustering algorithm. Typical examples of works include [205], [206], [207], and [208].

Relying on traditional methods for vehicle detection would be associated with many problems. First, hand-crafted features are time-consuming, and the hard threshold decision results are unsuitable for segmentation. Second, these methods cannot recognize a pixel-level object category as the same as semantic segmentation; thus, they can realize the segmentation but not pixel-wise classification. Therefore, these methods are not suitable for real-time vehicle detection scenarios. However, these methods can be adopted for point cloud pre-processing, increasing the accuracy of vehicle semantic segmentation based on deep learning methods.

Currently, the deep learning-based semantic segmentation method for vehicle detection has proven worth applying in this field. According to input data representation, these methods fall into point-based, voxel-based, projection-based, and hybrid-based categories.

a) *Raw data point cloud-based methods*: These methods do not modify the sparsity of point clouds; they directly extract features from raw point clouds and feed them to the model to detect vehicle targets. PointNet [157] and PointNet++ [158] models are pioneering examples of this category. Their model architectures use MLP and max-pooling to extract point-wise features and a global feature of the point set from irregular sparse, raw point cloud data. However, grouping local scale points through max-pooling results in low performance in the complex scene. SalsaNet [209] and S3Net [210] were proposed to solve this issue. In addition, PointNet adopts the sliding window to segment the point cloud, which may constrain the contextual information within the window size.

TABLE VI
THE LIDAR VEHICLE DETECTION PERFORMANCE FOR DIFFERENT MODELS ON THE KITTI TEST SET. THE RESULTS WERE EVALUATED BY THE MEAN AVERAGE PRECISION WITH 40 RECALL POSITIONS. MOD STANDS FOR MODALITY, L FOR LIDAR ONLY, LR FOR LIDAR+CAMERA

Detector		Mod.	Time (ms)	AP(IoU=0.7)			Year
				Easy	Med.	hard	
PointNet-based (Raw data)							
IPOD [169]		LR	-	79.75	72.57	66.33	2018
RoarNet [170]		LR	-	83.71	73.04	49.16	2018
PointRCNN [156]		L	260	85.94	75.94	68.32	2019
STD [171]		L	-	86.61	77.63	76.06	2019
Part-A2 [172]		L	-	85.94	77.95	72	2020
3DSSD [173]		L	38	88.36	79.57	74.55	2020
SASSD [174]		L	40.1	88.75	79.79	74.16	2020
CIA-SSD [175]		L	30.76	89.59	80.28	72.87	2020
Point-GNN [176]		L	643	88.3	79.47	72.29	2020
LSNet [177]		L	73	86.13	73.55	68.58	2021
Pyramid RCNN [178]		L	112.1	87.03	80.30	76.48	2021
ST3D [179]		L	-	-	-	74.61	2021
POAT-Net [180]		L	29	90.39	81.34	76.99	2021
SASA [181]		L	35.9	88.76	82.16	77.16	2022
PointDistiller [182]		L	-	88.1	76.9	73.8	2022
Projection-based							
DepthCN [159]		L	-	37.59	23.21	18.01	2017
MV3D [183]		LR	360	71.29	62.68	56.56	2017
RT3D [162]		L	89	72.85	61.64	64.38	2018
BirdNet [163]		L	110	88.92	67.56	68.59	2018
PIXOR [184]		L	93	81.7	77.05	72.95	2018
Complex-YOLO [185]		L	19.84	67.72	64.00	63.01	2018
BirdNet++ [186]		L	100	70.14	51.85	50.03	2020
MVL Net [187]		L	7	-	-	-	2020
RAANet [188]		L	62	-	-	-	2021
Pseudo-L [189]		LR	-	23.74	17.74	15.14	2022
SRDL [190]		LR	82	87.73	80.38	76.27	2022
Ri-Fusion [191]		LR	38.75	85.62	75.35	68.31	2023
Voxel-based							
3DFCN [154]		L	-	84.2	75.3	68.0	2017
VoxelNet [164]		L	33	77.47	65.11	57.73	2018
MVX-Net [192]		LR	-	72.85	61.64	64.48	2018
F-pointNet [193]		LR	-	88.92	67.56	68.59	2018
PVRCNN [166]		L	80	90.25	81.43	76.82	2020
SECOND [168]		L	50	83.13	73.66	66.20	2020
HVNet [194]		L	32	87.21	77.58	71.79	2020
Associate-3Ddet [195]		L	-	79.17	89.29	77.76	2020
VoxelRCNN [165]		L	40	90.09	81.62	77.06	2021
TANet [196]		L	34.75	83.81	75.38	67.66	2021
SIENet [197]		L	161.3	88.22	81.71	77.22	2022
SMSNet [198]		L	23.8	87.01	76.21	70.45	2022
MA-MFFC [199]		L	140	92.60	84.98	83.21	2022
PDV [200]		L	135	90.43	81.86	77.49	2022
SAT-GCN [201]		L	122	79.46	86.55	78.12	2023
Pillar-based							
PointPillars [202]		L	-	79.05	74.99	68.30	2019
ASCNet [203]		L	-	88.48	81.67	76.93	2022
WCNN3D [204]		L	-	88.57	78.04	76.16	2022

Therefore, many researchers proposed new models to solve the PointNet-associated challenges and problems. These include [211], [212], [213], [214], [215], [216], and [217], which are convolution point-based methods in which they extract the point-wise features by convolving the point with neighboring points using a correlation function. Presently, being inspired by the PointNet architecture, many researchers are improving the semantic segmentation performance. This includes the model in [218], where they adopt the super-point graph (SPG) that provides compact and rich context information and effectively enhances the performance of the

TABLE VII
THE PERFORMANCE COMPARISON FOR DIFFERENT SEGMENTATION
MODELS IN TERMS OF MEAN IOU, PRECISION,
AND OVERALL ACCURACY

Model	Dataset	mIoU	mAP	OA	Year
Raw Data-based					
PointNet [157]	S3DIS	47.71	66.2	78.62	2017
PointNet++ [158]	S3DIS	54.5	67.1	81	2017
SalsaNet [209]	KITTI	71.44	77.03	NA	2020
S3Net [210]	KITTI	66.8	NA	NA	2021
RandLaNet [213]	S3DIS	70.0	82	88	2020
KpConv [215]	S3DIS	70.6	79.1	NA	2019
ShellNet [217]	S3DIS	66.8	NA	87.1	2019
Projection-based					
PointSeg [219]	KITTI	67.4	748	NA	2018
SnapNet [220]	semanticKITTI	59.1	NA	88.6	2018
SqueezeSeg [160]	KITTI	66.34	70.7	NA	2018
SqueezeSegv2 [221]	KITTI	70.82	77.25	NA	2019
MVCNN [222]	SketchClean	NA	NA	87.2	2015
GVCNN [223]	ModelNet	NA	85.7	92.6	2018
Voxel-based					
VoxNet [224]	ModelNet40	NA	NA	83	2015
3DmFV [225]	ModelNet40	NA	NA	91.6	2018
SegCloud [226]	KITTI	36.87	NA	49.46	2017
OctNet [227]	RueMongue2014	59.2	73.6	81.5	2017
Kd-Net [228]	ShapeNet-core	82.3	NA	NA	2017
VIN [229]	nuScene	73.7	45.2	NA	2021
PointGrid [230]	ModelNet40	NA	NA	92	2018
O-CNN [231]	ModelNet40	NA	NA	89.9	2018
Hybrid-based					
SegVoxelNet [232]	KITTI	75.81	NA	NA	2020
2-S3Net [233]	semanticKITTI	69.7	NA	NA	2021
AMVNet [234]	semanticKITTI	65.3	NA	NA	2021
FusionNet [235]	semanticKITTI	61.3	NA	91.2	2020
JS3C-Net [236]	semanticKITTI	66.0	NA	NA	2021

point cloud segmentation. The performance comparison for different models is represented in Table VII.

b) Projection-based methods: The sparsity and unordered structure of the point cloud makes them unsuitable for direct feature extraction using CNN. Projection-based method project points cloud into 2D images and adopt 2D CNN for semantic segmentation of the projected image. Segmented 3D scenes are then obtained by reconstructing the 2D scene into the 3D scene. The projection is mainly BEV or FV projection. The typical examples of these methods include PointSeg [219], inspired by squeezeNet [107], SnapNet [220], and SqueezeSegV1-2 [160], [221]. These methods benefit from the existing well-developed 2D CNN algorithms making them fast in vehicle detection. However, projecting 3D point clouds to 2D data distorts some point clouds to 2D data, losing some information. Some scholars proposed a multi-view projection to improve the projection-based methods' performance.

Typical examples of these models are MVCNN [222] and GVCNN [223]. The multi-view method differs from ordinary projection since it observes the vehicle from different angles, making it more robust than standard projection methods.

c) Voxel-based methods: These methods transform the unstructured points cloud into a structured voxel of point cloud and apply 3DCNN to perform semantic segmentation. One benefit of the voxelization process is that it preserves the

depth information of the point cloud. However, to keep the original dimension of the point cloud, they have to sacrifice their resolution. Moreover, the performance of these methods heavily depends on voxel resolution [225].

Therefore, they inevitably have some limitations in their performance due to this voxel resolution dependency. Some pioneering voxel-based methods include VoxNet [224] and 3DmFV. And after that, many scholars have applied this voxelization technique in vehicle detection. Some exemplar models include SegCloud [226], OctNet [227], Kd-Net [228], VIN [229], and SPVNAS [237]. Voxel-based methods are computationally expensive. To cope with this issue, researchers remove the computations to the empty voxels, accelerating the inference speed.

Some research benefited from the combination of points and grids. PointGrid [230] was structured with points and grids where each grid cell uses simple points to quantify local features. In contrast, other researchers benefited from sparse octupole. O-CNN [231] convolves on a sparse octupole and uses the sparsity represented by octree and local direction of shape for useful memory allocation, reducing computation burden.

d) Hybrid-based method: Hybrid-based segmentation methods combine two or more previous techniques to extract features from point clouds. These can be a combination of voxel and projection-based methods, including BEV. Thus, we can combine voxel and point-based techniques or voxel, point, and projection-based techniques. In addition, the hybrid-based method combines different backbones by fusing them with the help of concatenation, attention, or add. Some illustrative examples include SegVoxelNet [232], 2-s3Net [233], AMVNet [234], FusionNet [235], and HySeg [238]. Hybrid models improve inference performance in terms of precision and accuracy. However, they require a lot of computation, making them not applicable in real scenarios.

Both projection- and voxel-based vehicle segmentation methods are associated with information loss resulting in low performance of the segmentation results in terms of precision and accuracy. Point cloud-based methods are associated with the considerable advantage of directly extracting features from the raw point cloud and preserving their data information. However, these methods are still computationally expensive due to the sparsity and unordered structure of the point cloud. Therefore, these methods are becoming a research trend in the autonomous driving industry. It is worth mentioning that some of these methods were originally not designed for vehicle detection. However, they can be retrained through generalization and apply them to vehicle detection.

C. mmWave Radar-Based Vehicle Detection Methods

MmWave radar transmits pulses of mm electromagnetic wave energy and detects vehicle targets from the reflections it receives. Facilitated by the antenna array processing, the mmWave radar can obtain the angular information of vehicle reflecting points, which, combined with the time of flight (ToF), can localize the points in 3D space. Compared with other sensors such as LiDAR and cameras, mmWave radar signals are less affected by severe weather

conditions (except heavy rain) and are insensitive to lighting. Moreover, these radars can provide accurate vehicle motion state and depth information, making them an excellent choice in autonomous driving perception (see Fig. S7 of the supplementary materials).

However, the form factor constraint of the antenna array may result in low radar resolution [239] and specular reflection [240], making the radar generate partial and sparse point clouds of the detected vehicles. Data representation of mmWave radar is usually a set of sparse detection points. A flow chart representing different processes to obtain radar data is shown in Fig. S8 of the supplementary materials and general vehicle detection processes chain is shown in Fig. 7. Depending on the signal output, mmWave-based vehicle detection methods can be classified into two categories. These are raw-data-level and map-level radar detection.

1) Raw Data Level Radar Detection: MmWave raw data contain the required information to locate the detected vehicles. Nowadays, with the development of radar technology, most radars, after receiving the reflected signal, can directly determine the detected target information such as range, velocity, angle, and reflection intensity of detected vehicles. Radar raw data level detection results can be classified as either moving or stationary targets. However, these raw data are sparser and noisier due to the low resolution of the radar [241], resulting in another class known as false detection.

Moving targets include dynamic vehicles, cyclists, and pedestrians, whereas stationary targets include parked cars, guard rails, trees, road curbs, and street lamps. The invalid targets due to interference or noise are then classified as false targets. The radar only extracts the raw data information of the detected target but cannot classify the detected targets. Some operations are then needed to classify moving vehicles to separate them into stationary and false targets. Research has shown that false target signals last a short time when detected. Currently, many approaches exist to eliminate these signals. Typical examples include iterative adaptive approach (IAA) [242], multiple hypothesis target (MHT) [243], and the Kalman filter [244].

Most known radar taxonomies include radar cross section (RCS) or radar signature and signal-to-noise ratio (SNR). SNR is known to be the measure of the strength of the desired signal relative to background noise, whereas the RCS is a measure of how detectable an object is by radar. Thus, the more prominent the RCS, the easier the object is detected. Research has shown that the RCS and SNR of stationary targets are small compared to moving vehicles. Therefore, by setting a practicable threshold for RCS and SNR of the radar, the moving vehicle target can be separated from stationary ones. It can be mentioned that recognition and classification can be performed by applying some ML algorithms such as support vector machine (SVM) or neural network (NN). In the IVs' environmental perception, raw data target detection plays a vital role due to its inherent measurement of the vehicle targets' relative velocity and position. However, it is unable to provide the vehicle types and their size. Moreover, the radar detection accuracy is not optimal when the vehicle is stationary or moving at a low speed. Therefore, relying on the mmWave radar as the only IV sensor is unsuitable.

2) Map-Level Radar Target Detection: Compared to raw data target detection radar, the imaging radar not only provides the target's velocity and motion state but also outputs the radar signals into image maps. Different categories of output format have been proposed. And they can be categorized into two groups which are 2D grid maps and 3D point cloud maps. Some examples of radar image-level detection output formats are represented in Fig. S9 of the supplementary materials.

a) 2D grid maps: 2D representation can be classified as range-doppler-azimuth maps and projections maps. On the first hand, we have the range-doppler-azimuth maps category. This category is considered when we use a range fast-Fourier transform (FFT) profile, azimuth elevation stack, range angle, or doppler signature to express critical radar information such as range, doppler, or azimuth. On the other hand, when the reflection intensity of the mmWave radar detection target is projected onto the image to form the reflection intensity map, perspective map, or BEV map, they fall into the projection maps category. 2D representation maps are viral since most modern deep learning algorithms, such as CNN, FCN, and long short-term memory networks (LSTM), can be easily adapted to these representations.

b) 3D point cloud maps: 3D points cloud preserves all the information by sacrificing the cost of higher sparsity. Point clouds provide the spatial data of each point [245], [246]. To detect vehicles, clusterization is used to enrich the information of each point, and ML or DL algorithms are applied for classification. Authors in [247] clustered the points cloud by the DBSCAN and applied the SVM to classify the vehicle target. Due to the maturity of 2D modern DL algorithms, the point cloud can be quantized into discrete voxel grid heatmaps so that DL algorithms can be used [248]. A discretized space allows CNN techniques to be used. However, the sparsity of the point cloud and the low resolution of the radar create a trade-off for getting better performance.

To improve the performance, authors in [249] adopted the GAN network to cope with the resolution issues and applied segmentation techniques based on CNN for vehicle detection. Scholars in [250] enhance the capacity of automotive radar by exploiting the temporal information from successive ego-centric bird-eye-view radar image frames for radar object recognition. They suggest a temporal relational layer to explicitly represent the relationships between objects inside subsequent radar images by taking advantage of the consistency of an object's presence and attributes (size, orientation, etc.).

With the development of synthetic aperture technology, synthetic aperture radar (SAR) deals with low-resolution issues by combining the doppler and synthetic aperture technology. Some models like MilliPoint [251] present a self-tracking algorithm with mm precision and enable SAR imaging with finer resolution achieving 3D point cloud generation for IV applications. The SAR outputs the spherical image known as the SAR map, which benefits from all the state-of-the-art DL target detection algorithms such as CNN, SSD, YOLO, and semantic segmentation.

The use of mmWave radars has recently attracted more interest in autonomous driving thanks to its advantages against other sensors. They are cheaper and mostly unaffected by

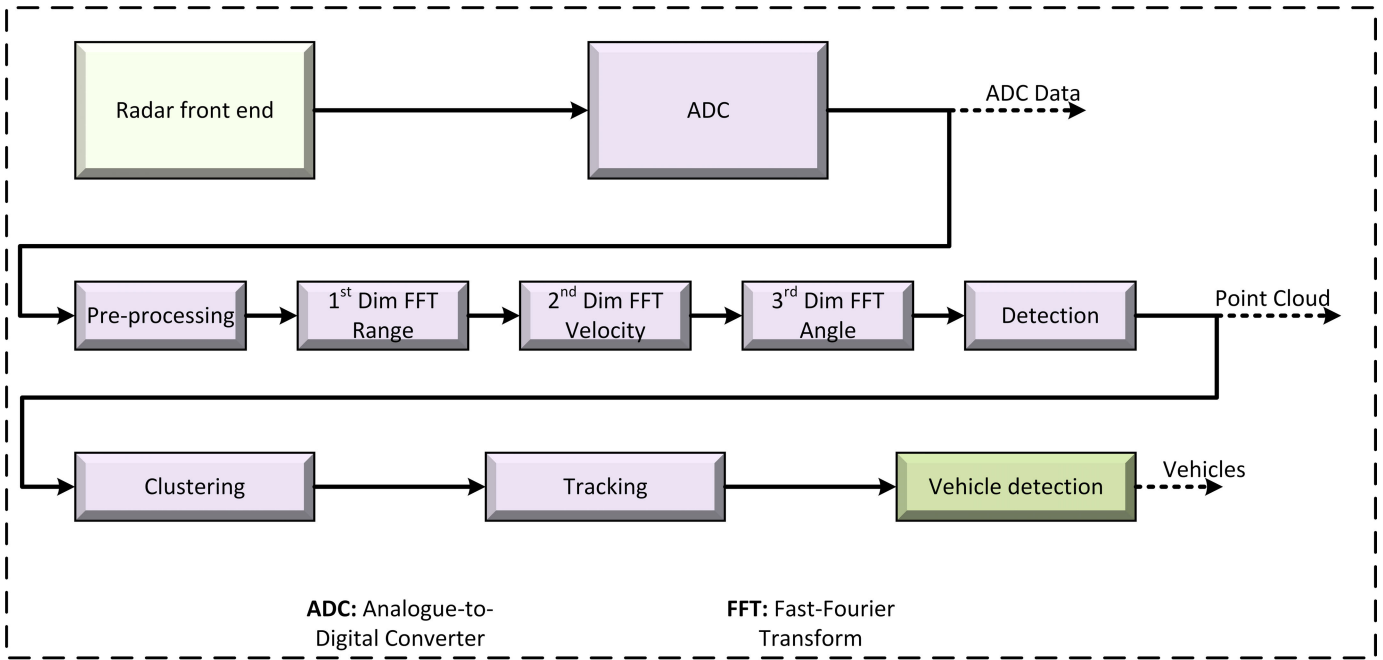


Fig. 7. General processes for vehicle detection based on radar.

adverse weather and smoke. However, their spatial resolution is not good compared to LiDAR; Moreover, they are more sensitive to noise and interference. Therefore, fusing this sensor with other sensors is reasonable for a better environment perception for IVs. Currently, more research is being carried out to improve the mmWave radar resolution, and with the development of radar technology, these radars will be widely used in the autonomous driving industry.

D. Multi-Sensor Fusion-Based Method for Vehicle Detection

The information from a single sensor cannot guarantee the safety requirement and the degree of accuracy required for an intelligent vehicle in complex autonomous environment perception. Every sensor has its pros and cons in terms of environment perception. Table VIII compares the characteristics of three sensors (camera, mmWave radar, and LiDAR). As we mentioned, these three sensors have complementary advantages. Therefore, multi-sensor fusion is an inevitable topic in autonomous driving. The advantages and disadvantages of standard sensor fusion techniques are summarized in the Fig. S10 of the supplementary materials.

1) *Stereo Vision-Based Vehicle Detection Methods*: In the stereo vision system, vehicle detection is performed on information obtained from a combination of two or more cameras capturing different perspectives of the given scene. The vehicle's accurate position and spatial depth information are obtained by searching for point correspondences in the disparity map of the two images. Based on the methodology principle, the stereovision method can be classified as appearance-based or motion-based.

a) *Vehicle detection appearance-based*: These methods rely on extracting the salient features to detect the vehicle targets. Authors in [252] considered the symmetrical appearance of the vehicle as their main way to detect vehicle targets. Generally, the vehicle's rears are symmetrically

rectangular with a specific aspect ratio constraint. Thus, vertical feature extraction followed by a matching process based on symmetry is adopted to detect the vehicles. Authors in [253] adopt a feature extraction stereo algorithm, followed by epipolar constraint and feature matching based on the normalized cross-correlation (NCC) to detect vehicle targets.

Researchers in [254] benefited from the fact that the depth information of the vehicle changes continuously and adopts depth information clustering to detect the vehicle. Some scholars used ML algorithms to train vehicle detection classifiers. For example, authors in [255] adopted a block-matching algorithm which compares the sum of absolute differences (SAD) between two blocks of pixels from an image to get the disparity map. The clustering algorithm based on the disjoint-set data structure [256] is used, and HOG combined with SVM was adopted to train the vehicle classifier.

b) *Vehicle detection motion-based methods*: These methods benefit from the optical flow information. Authors in [257] estimated the change in the optical flow features of two close frames to track the target's position. Some scholars, such as [257], clustered the areas with remarkable optical flow features. Whereas other researchers like [258] merged the depth information of the vehicles to their optical flow by using an occupancy grid, which highly improved vehicle detection efficiency.

2) Vehicle Detection Based on the Fusion of mmWave

Radar and Vision: The mmWave radar and the cameras are the classical vehicular sensors commonly used for vehicle detection and tracking. The mmWave radar is known to provide depth information and the motion state of the vehicle. In contrast, the camera image is known to have a high resolution containing rich semantic information. Despite their promising abilities, their drawbacks are insufficient for environmental perception when used separately. Therefore, fusing mmWave radar and the camera compliments each

TABLE VIII

COMPARISON BETWEEN DIFFERENT EXISTING SENSORS. “1”-“6” REPRESENTS THE LEVELS FROM “EXTREMELY LOW” TO “EXTREMELY HIGH”. AR STANDS FOR ANGLE RESOLUTION, RR: RANGE RESOLUTION, OD: OBJECT DETECTION, OC: OBJECT CLASSIFICATION, FOV: FIELD OF VIEW, LS: LIGHT SENSITIVITY, CP: COLOR PERCEPTION, AND RWC: REQUIREMENT FOR WEATHER CONDITION

Sensor	AR	RR	OD	OC	FOV	LS	CP	RWC	Cost	Size
Camera	6	2	6	6	3	1	6	4	3	2
Radar	4	4	5	1	4	1	1	1	2	1
LiDAR	5	5	5	3	4	1	1	4	5	5

other for better environment perception of the ego vehicle. Generally, mmWave radar and camera fusion-based vehicle detection typically fall into three classes. These include data-level fusion, feature-level fusion, and decision-level fusion as shown in Fig. 8. Table IX summarizes pros and cons of radar camera fusion level.

a) Data level fusion: Data level fusion was a classic environment perception scheme, but it is no longer a research trend. However, its idea of fusing radar and camera data to proceed with the target represents the data-level fusion process. Generally, data-level fusion methods first use the radar points to generate ROI [259]. And the image-based vehicle detection method is used to generate vehicle targets from the ROI, and finally, a feature extractor and classifier are adopted to detect vehicle targets.

For example, scholars in [260] use the mmWave radar to detect on-road vehicles and transmit the ROI size and location to an image sequence captured by a monocular camera. Then, the vision module generates a square box around the ROI in the image, and the on-road vehicles are detected with the help of the active contour method. Some scholars have adopted the DL to perform vehicle detection. YOLO [65] and SSD [70] based models are the most commonly used to perform the detection. CramNet [261] is a camera-radar fusion with ray-constrained cross-attention for robust 3D object detection, where the ray-constrained cross-attention technique resolves the ambiguity in the geometric correspondences between camera features and radar features. It can be mentioned that the adequate number of radar points directly impacts the vehicle detection results. Thus, if there is no radar point for a particular area of the image, that area will be disregarded.

These will save some computation costs. However, the worst scenario may happen if the radar points are not well processed. The general architecture of data-level fusion for vehicle detection is shown in Fig. 8a.

b) Decision level fusion: The decision level fusion fuses the final processing results of camera and radar sensors. The camera is associated with a large field of view (FOV), whereas the mmWave radar is associated with a tremendous longitudinal FOV. Therefore, the decision fusion level benefits from both sensor information to detect vehicle targets.

However, the radar noise differs from the camera noise, resulting in a challenging task when designing a filtering technique, especially the joint probability density function. To achieve radar-vision decision-level fusion, scholars [262] adopted Bayesian theory, [263] adopted Kalman filter, [264] adopted Dempster Shafer theory whereas [265] based on radar validation to fuse the detection results from both camera and radar. The general architecture of existing radar-camera

decision-level fusion techniques for vehicle target detection is presented in Fig. 8b.

c) Feature level fusion: Feature level fusion is an environment perception scheme in which the extracted features from both sensors (radar, camera) are fused to detect vehicle targets [266], [267], [268], [269], [270]. The feature maps from the camera image can be extracted using conventional machine vision methods. The radar point feature can also be extracted based on transforming the radar point into image form. After getting the independent feature maps for each sensor, a CNN-based model can be applied to simultaneously learn the camera and the radar features realizing the feature fusion.

Some authors adopted more image feature extractors to improve the model's efficiency. FusionNet [271] adopted SSD to generate feature maps followed by detection heads. However, instead of using one feature extraction network for camera images, as in reference [235], they combined input from different sources observing the same physical scene. Authors in [272] contribute to radar-camera feature level fusion by bridging the view disparity between radar and camera features for multi-modal fusion 3D object detection. The general architecture of feature-level fusion for vehicle detection is shown in Fig. 8c.

3) LiDAR and Vision Fusion-Based Vehicle Detection: LiDAR vision-based vehicle detection methodologies can be classified into two classes based on different fusion stages. These are early fusion and late fusion. Early fusion can be categorized as data-level fusion and feature-level fusion, whereas late fusion can be classified as decision-level fusion and multi-level fusion, as shown in Fig. 9.

a) Early fusion methods: The early fusion methods first fuse the original data or features of the two sensors, thus LiDAR and camera. And it can be categorized as data-level fusion and feature-level fusion.

Data-Level Fusion: fusing the data of two different sensors seems complicated since the structure of these sensor data is different. Moreover, if combined, it can result in the anti-interference performance of the model leading to failure in detecting vehicle targets [273]. Therefore, this fusion level is ideal for fusing similar data, thus from similar sensors.

Feature Level Fusion: To solve the difficulties in fusing data of different natures, the LiDAR point cloud features and camera image features are first extracted and merged with some fusion algorithms since they have the same data structure. Many researchers used these techniques to fuse LiDAR and vision. Authors in [274] combine LiDAR and camera RGB images to detect cars, cyclists and pedestrians. First, the RGB image feature maps are extracted based on the residual modules. Second, the LiDAR points cloud is

TABLE IX
ADVANTAGES AND DISADVANTAGES OF RADAR CAMERA FUSION LEVELS

Fusion Level	Pros	Cons
Data level fusion	Reliable and low data loss	Highly affected by the number of radar points
Decision level fusion	Fully take advantage of sensor information.	The implementation of the joint probability density function of sensors is tough.
Feature level fusion	Fully utilize feature information resulting in high detection performance ability.	Sacrifice computation resources and overhead of radar information transformation.

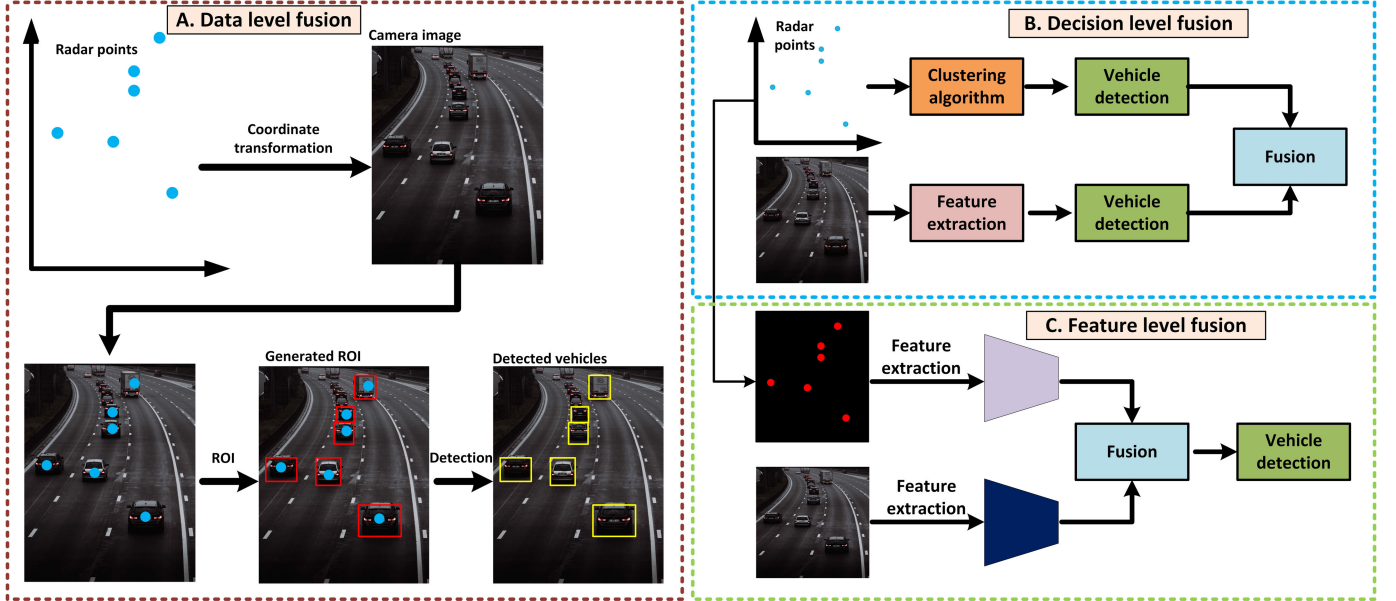


Fig. 8. The general architecture of different radar camera fusion for vehicle detection (a,b,c).

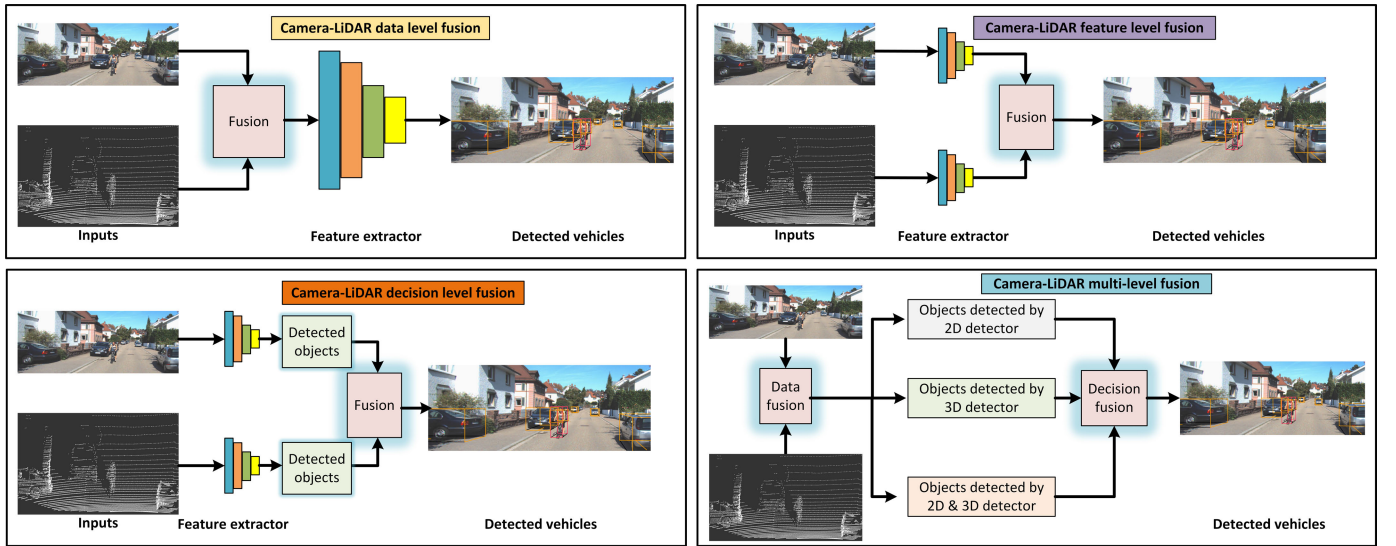


Fig. 9. The general architecture of different LiDAR-camera fusion for vehicle detection.

transformed into BEV, and the BEV feature maps are extracted based on sparse convolution. And finally, both feature maps are fused and fed to RPN to obtain proposals for object classification and the bounding box regression.

Researchers in [275] considered that the geometrical consistency in the local 3D and 2D is helpful for regression tasks in 3D object detection and proposed a 3D-2D consistent feature. The hand-crafted 2D features and the attentional

structure voxel feature are fused to generate the primary semantic features for 3D classification and detection tasks.

Authors from the Baidu laboratory [183] took the FV, the BEV of the LiDAR, and the camera image as the input. The team first adopts the BEV map to generate 3d proposals and then projects them into a three view. The ROI pooling on each view was performed to obtain the region-wise feature maps, which were then fused by a deep

fusion algorithm to perform 3D classification and detection tasks.

Authors in [276] first project the LiDAR point cloud information into grids to detect targets, and then these targets are mapped into the image to get ROIs. The dynamic threshold is used to expand the obtained ROIs, which are fused to get the final ROIs. Then the YOLO is used to detect the vehicles.

The work in [277] proposes a point feature fusion module that directly extracts point-wise features from raw RGB images and fuses them with their corresponding point cloud with no backbone. The network novelty is that they abandoned the backbone, which extracts RGB feature maps to minimize computational costs. Reference [278] use 3D LiDAR points to effectively generate accurate object region proposals. These candidates are mapped into image space where the ROIs are selected and fed to CNN for vehicle detection.

The feature level fusion preserves the sensor data's original features, dramatically improving the vehicle detector's performance. However, extracting feature maps from multiple sensor data results in weak anti-interference performance, reducing the efficiency.

b) Late fusion methods: The late fusion methods fuse the final processing results of each sensor (camera and LiDAR). The late fusion method can be categorized as decision-level and multi-level fusion.

Decision Level Fusion: This method first adopts different ways to detect vehicle targets from each data type, thus LiDAR point cloud and image. And then, fusion techniques are used to fuse the final processing results of those sensors. The representative fusion algorithms include the Kalman filter, Bayesian and evidence theory. Many scholars adopted decision level fusion for target detection. For example, authors in [273] proposed a depth completion method to convert a 2D LiDAR sparse depth map into a dense depth map so that they can align two sensors with each other at the level of data.

Then YOLOV3 [67] was used for vehicle target detection from RGB images and depth maps. And finally, a decision-level fusion based on a bounding box and improved Dempster-Shafer (D-S) evidence theory was used to fuse the two results and obtain final vehicle position and distance information. Authors in [279] improved YOLO for dim target detection, and then a decision-level fusion was adopted to fuse the RGB image and the depth image for accuracy improvements. Their network does not consider the light intensity, which is a big challenge for the camera.

Since both LiDAR and stereo vision can get depth information of the vehicle targets, fusing the information from both sensors at the decision level seems 'easy and accurate'. In addition, the decision level has a strong anti-interference performance compared to feature fusion level methods. This method cannot only avoid system failure caused by conflicting sensors but also run when one of the sensors fails.

Multi-Level Fusion: The author in [280] developed a three stages network for LiDAR and camera fusion 3D vehicle detection. The network involves data level fusion of points clouds and images, feature level fusion of voxels and BEV in the point cloud branch, and the feature level fusion of point cloud and images. In addition, a coarse fine header was proposed to stimulate two-stage detectors, which generate

rough proposals on the encoder and refine them in the decoder. Generally, a camera can project a remote vehicle target, which is challenging for a LiDAR. Thus, a 2D camera can initialize the tracking when the object is far away, but LiDAR cannot unless the object is close. Some researchers exploited this gap between LiDAR and the camera.

The authors in [281] proposed a 4-level deep association mechanism that realizes tracking of an object in the 2D domain when the object is far and can only be detected by the camera and updating 2D trajectory with 3D information obtained when the object is in the LiDAR FOV to achieve a smooth fusion of 2D and 3D trajectories. Thus, 3D bounding boxes from LiDAR are projected to 2D bounding boxes from the camera, and their IoU is calculated. Then, a threshold is set to achieve the information fusion from both sensors. Some researchers like in [282], adopt sensor calibration and coordinate transformation techniques to project the LiDAR point cloud onto the image. The vision vehicle detection algorithm is used to verify the hypothesis of the ROI.

4) Multi-Modalities Fusion: Vehicle detection is an autonomous vehicle's most critical environment perception task. Note that perception sensing is collecting data information from different vehicle sensors and processing these data for an ego car to understand its surroundings. Therefore, information from a single sensor cannot guarantee the reliability of cognitive technology in the complex autonomous vehicle environment.

The above sections have detailed sensor fusion techniques for vehicle detection, including stereo-vision-based, LiDAR-vision-based, and radar-vision-based. Multi-modalities fusion involves fusing more than one camera, LiDARs, and radars into one intelligent vehicle's perception system to accomplish perception tasks. According to the degree of information abstraction, multi-modalities fusion typically falls into these two categories: early fusion, which includes data-and feature-level fusion, and late fusion, which consists of decision-level fusion and multi-level fusion.

The multi-modalities fusion is still in the exploratory stages. However, the results from different researchers, such as [283] and [284], present better environment perception than the existing methods. With the maturity of the sensor, there is no doubt that multi-sensor fusion will be a hot topic in the field of vehicle detection.

IV. PERFORMANCE ANALYSIS, CHALLENGES, AND FUTURE TRENDS

Sections II and III of this work provided a synopsis of algorithms and datasets for vehicle detection in autonomous driving. Thus, we outlined the existing datasets and methodologies for vehicle detection regarding environment perception sensors like cameras, LiDAR, and mmWave radar. In this section, we examine the environment perception sensors' performances for vehicle detection and explore some existing challenges and future trends of these vehicle detectors typically, in the field of autonomous driving.

A. Sensor's Performance Analysis

1) Camera: Currently, vehicle detection methods based on machine vision have significantly impacted the autonomous

driving industry since they benefit from well-developed and well-matured 2D machine vision deep learning-based algorithms. These methods are known to be the best feature extractors, making them the best choice for the vehicle or object detection. Moreover, deep learning-based detectors have proven incomparable to traditional machine vision detectors' speed and accuracy.

Generally, deep learning-based vehicle detectors are categorized into one- and two-stage networks. The two-stage network detectors are more accurate but low in speed compared to the one-stage network or vice-versa. However, both networks are anchor-based networks that require complex anchor parameter tuning followed by computationally expensive post-processing, making them inefficient. Nowadays, anchor-free networks' performance and efficiency surpass state-of-art anchor-based networks. CenterNet [78] is the representative anchor-free network representing objects as points. It has opened the bridge for other researchers to develop more anchor-free networks.

It can be mentioned that vehicle detection is associated with inevitable real-time performance requirements to fulfill a man's wish for an IV. Therefore, researchers have adopted anchor-free networks in IVs for increased performance. Illustrative example networks include CenterNet3D [285], 3D-DFM [286], TBox [287], FII-CenterNet [288], 3D-CenterNet [289], [290], and [291]. These networks have shown significant improvement in terms of speed and accuracy. Therefore, there is no doubt that researchers will focus on one-stage anchor-free vehicle detector networks in the future.

2) *LiDAR*: An active sensor is often used in intelligent vehicles due to its undoubtedly significant advantages of being insensitive to light conditions and obtaining the ego vehicle's surrounding 3D information at very high precision. When using DL techniques, these sensors achieved significant performance over traditional methods for vehicle detection in terms of accuracy and speed. However, the sparseness and unstructured form of LiDAR point cloud data bring the inevitable computational cost to process them in their natural state.

Therefore, many researchers prefer traditional methods to pre-process the raw data point cloud and then apply DL for vehicle detection. These reduce the computational cost by sacrificing some raw point cloud information which may result in low performance. To cope with this issue, researchers have started to design vehicle detectors relying on raw point cloud data. The research is still in the exploratory stages; however, preliminary results proved it is possible. Hence, it is hoped to get a compact vehicle detector that expeditiously detects vehicles from raw data points clouds.

3) *mmWave Radar*: One of the inexpensive environment perception sensors in the autonomous driving field, it comes with the ability to perform all-times, all-weather while sensing the precise distance and motion of the vehicle. However, their resolution is relatively low, making them unable to provide vehicle type and size, which is an essential task in vehicle detection. The fusion of mmWave radar with another sensor solves this issue.

Currently, most mmWave radar-based methodologies first use radar information to generate ROI, and deep-learning

methods are used to detect vehicles in those ROIs, significantly improving vehicle detection accuracy. The application of mmWave radar is still in the exploratory stage.

However, the all-weather performance ability makes the radar an inevitable topic in autonomous vehicles, especially when detecting vehicles under adverse weather conditions. Therefore, it is expected that with the development of radar technology, the application of radar-based vehicle detectors will be a trending topic for intelligent vehicle research.

B. Challenges

1) *Real-Time Performance*: Deep learning-based models for vehicle detection have a higher performance than other computer vision algorithms. However, the realization of the real-time performance of these models is still a big challenge. Vehicle detectors need to be expeditious in terms of detection speed and accuracy to meet real-time performance requirements.

Currently, existing approaches are either designed with complicated network structures, which result in high accuracy but slow detection speed, or with simple, lightweight networks, which achieve a higher detection speed but slow detection accuracy. Thus, most of today's algorithms sacrifice the detection speed to improve the detection accuracy or vice-versa, which is insignificant.

Therefore, it is hoped that a compact and lightweight network will be designed to enhance vehicle detection accuracy and speed. Such lightweight, compact, accurate, and speedy models can be deployed into real-time scenes, realizing IVs' real-time performance.

2) *Datasets*: Deep learning-based models are known to be data-hungry models. They require massive data for training, which dramatically impacts the model performance. Most IVs utilize deep-learning algorithms for information extraction and decision-making. Therefore, they need enormous visual sensory data to be aware of their surroundings [292].

With the development of the AV industry, some companies and institutions provided large-scale datasets (e.g., KITTI), which enable researchers to study self-driving and aim to promote advanced research and development in autonomous driving and machine learning. However, the existing data seem insufficient to fit the deep learning models' needs.

Some data augmentation techniques are applied, and simulation environment data are used to increase the number of datasets for training. However, the real-scene datasets are still not enough to realize real-time performance. Some datasets are made publicly available by their owners while others are not or set some limitations for using their resources; for example, KITTI updated their data usage policy and limited the model result comparison for everyone. Only works to be published in peer-reviewed journals or conferences are allowed for results comparison.

Collecting and annotating datasets for adverse conditions remain as challenging problem in the AV industry. However, some researchers, like [292] and [293], have started to work on this issue. There is no doubt that adverse condition datasets will attract many companies and institutions.

Therefore, we hope that institutions should keep gathering more datasets and updating the existing ones to meet the

requirement of training those deep learning models. Moreover, we hope that more companies shall make their datasets publicly available for everyone to promote the advanced research and development of IVs.

3) *Diverse Weather Environment*: The vehicle detectors perfectly detect vehicles when working in a good traffic environment. However, the harsh traffic environments are still challenging for these approaches. Vehicle detection in a complex environment such as night, weak illumination, or adverse weather such as rainy, snowy, and foggy days is still ambitious. Since different sensors have complementary advantages, we hope that future research should focus on fusing these sensors' data to exploit the best of each sensor, which can improve the accuracy and robustness of vehicle detection in adverse weather.

IVs face some challenges due to immaturity in sensor technology, and we hope these issues will be resolved with the development of sensor technology. As an illustration, increasing the transmission frequency will resolve low-resolution problems associated with the radar. Weather denoising can be performed by different works proposed in [294]. Adding a laser beam to the LiDAR can improve its perception ability, or adopting a thermal infrared camera can work better under foggy days, improving vehicle perception in harsh environments. Thus, future technological changes in sensor technology will surely improve vehicle detection.

C. Future Trends

1) *Sensor Fusion*: As one of the most challenging tasks in the field of IVs, vehicle detection is performed by combining some computer algorithms and the data perceived from smart vehicle sensors. With the development of sensor technology, sensors have shown uprising support in IV detection. However, despite each sensor's promising potential, their drawbacks make them insufficient for vehicle detection when used separately.

Therefore, fusing multiple sensors to complement each other is a hot future topic. Researchers in [295] developed a detachable and expansible multi-sensor data fusion model. Their model has front fusion, global fusion, and a synthesizer. The model can be disassembled and configured for different trim levels of vehicles. Researchers shall design a fusion scheme that effectively fuses the sensor data information and efficiently meets real-time performance requirements.

2) *Multi-Driving Scene Vehicle Detector*: Most existing vehicle detection approaches carry their experiments under suitable traffic environments. Moreover, developing an algorithm capable of providing adequate performance in all driving scenarios is cumbersome [296], [297], [298]. Thus, only a few techniques design their model to cope with complex traffic environments such as night, weak illumination, rainy, snowy, and foggy days.

Therefore, vehicle detection in adverse weather is an inevitable trend in the future. Additionally, the existing models are prone to change in driving scene conditions. Thus, they reduce their performance when being experimented on in situations they were not meant to be. Therefore, future research should design a vehicle detector scheme that would

adapt to the varying traffic environment scene and increase the vehicle detection accuracy and speed to realize real-time applications.

3) *Cooperative Perception*: As a key component of intelligent connected vehicles, cooperative perception aims to use wireless communication technology to interact and combine local perception data with environmental data collected by edge nodes. On the first hand, this can decrease latency, eliminate system blind spots, and improve vehicle perception accuracy. Due to movement, the vehicle's perceptual system cannot continuously monitor its surroundings. Besides, onboard cameras have a relatively small field of view and are susceptible to obstruction.

On the other hand, since they are taken from cameras located a few meters above the ground, data from roadside cameras offer intrinsic advantages in terms of robustness against occlusion and long-term event prediction. Cooperative perception can be classified as vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I). Both methods enhance accuracy, improve the perception range, and reduce delay.

Additionally, this will give autonomous vehicles richer, more accurate real-time environmental data, and will establish the groundwork for implementing AD. For this reason, the number of onboard ego sensors can be decreased, reducing the price of AD, and high-level AD development can happen more quickly, thanks to cooperative perception. Some scholars have started to apply this technology to 3D object detection; this includes [298], who developed a monocular 3D vehicle localization network for roadside surveillance cameras named CenterLoc3D, a 3D vehicle detection without 2D detectors in roadside scenes.

P2OD [299] is a lightweight, privacy-preserving object detection network for connected autonomous vehicles. PillarGrid [300] is a novel cooperative perception method fusing information from multiple 3D LiDARs to improve connected and automated vehicle situation awareness. The Rope3D [27] is the first comprehensive dataset for roadside perception, with a view to advancing cooperative perception for AV. Therefore, future research should focus on cooperative perception to explore its benefits.

4) *Unsupervised Learning*: Generally, fully supervised deep learning-based models face three challenges. First, these models are data-hungry-driven, and the insufficient data to feed them for training is becoming a considerable challenge. Second, these models require annotated data for their model training which is arduous and time-consuming. And finally, the generalization ability for supervised learning-based models is not good, primarily when the model detects untrained objects. Therefore, research on weakly-supervised or fully unsupervised learning-based models should be necessary to solve the earlier challenges.

V. CONCLUSION

Vehicle detection is one of the most crucial tasks for autonomous driving. In this article, we comprehensively reviewed the existing methods and datasets for vehicle detection considering their performances and applications in the field of autonomous driving. First, we described tasks, evaluation criteria, and existing public datasets for vehicle

detection in autonomous driving. Second, we provided a rigorous review of the recent vehicle detection methods. Considering the kinds of sensors they used as a reference, the existing methods were categorized into machine vision-based, mmWave radar-based, LiDAR-based, and sensor fusion-based methods. Finally, we analyzed the pertinent challenges of AVs and provided recommendations for future works concerning vehicle detection.

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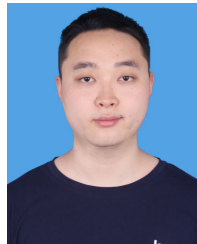
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